

A Rigorous Galerkin Closure for the Elastodynamic Volume Integral Equation on a Cube 27-Component Path-B Formulation

Research Notes

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1 Motivation: differentiating under the integral fails

The classical moment-method closure for the cube T-matrix begins from the Lippmann–Schwinger (LS) equation

$$u_i(\mathbf{r}) = u_i^0(\mathbf{r}) + \omega^2 \int_V G_{ij}(\mathbf{r} - \mathbf{r}') \delta\rho(\mathbf{r}') u_j(\mathbf{r}') dV' - \int_V \partial'_k G_{in}(\mathbf{r} - \mathbf{r}') \delta c_{nklj}(\mathbf{r}') \partial'_l u_j(\mathbf{r}') dV', \quad (1)$$

where G_{ij} is the elastostatic Green’s tensor and $V \subset \mathbb{R}^3$ is the inclusion. To obtain a closed finite-dimensional system one then formally takes derivatives at the centre $\mathbf{r} = \mathbf{0}$:

$$\begin{aligned} u_i(\mathbf{0}) &= u_i^0(\mathbf{0}) + \omega^2 \int_V G_{ij} \delta\rho u_j dV' - \int_V \partial'_k G_{in} \delta c_{nklj} \partial'_l u_j dV', \\ \partial_p u_i(\mathbf{0}) &= \partial_p u_i^0(\mathbf{0}) - \omega^2 \int_V \partial'_p G_{ij} \delta\rho u_j dV' + \int_V \partial'_p \partial'_k G_{in} \delta c_{nklj} \partial'_l u_j dV', \\ \partial_q \partial_p u_i(\mathbf{0}) &= \partial_q \partial_p u_i^0(\mathbf{0}) + \omega^2 \int_V \partial'_q \partial'_p G_{ij} \delta\rho u_j dV' - \int_V \partial'_q \partial'_p \partial'_k G_{in} \delta c_{nklj} \partial'_l u_j dV', \end{aligned} \quad (2)$$

where the r -derivatives have been converted to r' -derivatives by translation invariance $\partial_r G(\mathbf{r} - \mathbf{r}') = -\partial_{r'} G(\mathbf{r} - \mathbf{r}')$.

The Leibniz rule used in passing from (1) to (2) requires (DCT version) a dominating function $|\partial_r f(\mathbf{r}, \mathbf{r}')| \leq g(\mathbf{r}')$ with $g \in L^1(V)$. The static elastic kernel has the singularity

$$G_{ij}^{\text{stat}}(\mathbf{s}) \sim \frac{1}{|\mathbf{s}|}, \quad \partial' G \sim \frac{1}{|\mathbf{s}|^2}, \quad \partial' \partial' G \sim \frac{1}{|\mathbf{s}|^3}, \quad \partial' \partial' \partial' G \sim \frac{1}{|\mathbf{s}|^4}, \quad (3)$$

so already at the level of $\partial' G$ *no integrable dominating function exists* when $\mathbf{r}$ is in the support of the inclusion. Each of the equations in (2) therefore makes only *formal* sense; converting them into rigorous statements requires distributional regularization (Cauchy principal value, Hadamard finite part, and Eshelby-type delta corrections).

LS derivative order	worst kernel	status
0 (body force)	$1/ \mathbf{s} $	$L_{\text{loc}}^1 \checkmark$
0 (elastic kernel)	$1/ \mathbf{s} ^2$	conditionally convergent (PV)
1 (body force)	$1/ \mathbf{s} ^2$	PV
1 (elastic kernel)	$1/ \mathbf{s} ^3$	Eshelby delta required
2 (body force)	$1/ \mathbf{s} ^3$	Eshelby delta required
2 (elastic kernel)	$1/ \mathbf{s} ^4$	derivative-of-delta / Hadamard PF

Table 1: Singular structure of the formal closure equations (2).

2 Path B: Galerkin formulation on a polynomial trial space

We replace point evaluation by a Galerkin (weak) projection onto a finite-dimensional polynomial trial space. The resulting bilinear forms involve only G_{ij} itself (the singularity $1/|\mathbf{s}|$ is locally L^1 in $\mathbb{R}^3$), so *every entry of the closure matrix is an absolutely convergent integral*. No principal values, no delta corrections, no distributional bookkeeping.

2.1 Trial space T_{27}

Let $V = [-a, a]^3$. Denote by $P_2(V)$ the space of real polynomials in $\mathbf{r} = (r_1, r_2, r_3)$ of total degree ≤ 2 , $\dim P_2(V) = 10$. The naive vector trial space $(P_2(V))^3$ has dimension 30. We remove the 3 rigid-rotation modes by symmetrizing the linear part, giving the 27-dimensional subspace

$$T_{27} = \left\{ \boldsymbol{\varphi}(\mathbf{r}) \mid \varphi_i(\mathbf{r}) = U_i + \varepsilon_{ai} r_a + \frac{1}{2} M_{abi} r_a r_b, \quad \varepsilon_{ai} = \varepsilon_{ia}, \quad M_{abi} = M_{bai} \right\}. \quad (4)$$

The dimensions add as $3 + 6 + 18 = 27$, matching one displacement monopole, the symmetric strain tensor, and the symmetric second-derivative tensor of $\mathbf{u}$ (equivalent to the strain gradient $\partial_k \varepsilon_{ij}$ through the kinematic identity $\partial_k \partial_l u_m = \partial_k \varepsilon_{lm} + \partial_l \varepsilon_{km} - \partial_m \varepsilon_{kl}$).

2.2 Galerkin LS

Definition 1 (Path-B closure). The Path-B 27-component closure is the variational problem: find $\mathbf{u}^h \in T_{27}$ such that for all $\boldsymbol{\varphi} \in T_{27}$,

$$\langle \boldsymbol{\varphi}, \mathbf{u}^h \rangle_V = \langle \boldsymbol{\varphi}, \mathbf{u}^0 \rangle_V + \omega^2 \delta \rho \mathcal{B}_{\text{body}}[\boldsymbol{\varphi}, \mathbf{u}^h] - \delta c_{nklj} \mathcal{B}_{\text{el}}[\boldsymbol{\varphi}, \mathbf{u}^h], \quad (5)$$

where the bilinear forms are

$$\langle \boldsymbol{\varphi}, \mathbf{u} \rangle_V := \int_V \varphi_i(\mathbf{r}) u_i(\mathbf{r}) dV, \quad (6)$$

$$\mathcal{B}_{\text{body}}[\boldsymbol{\varphi}, \mathbf{u}] := \int_V \int_V \varphi_i(\mathbf{r}) G_{in}(\mathbf{r} - \mathbf{r}') u_n(\mathbf{r}') dV dV', \quad (7)$$

$$\mathcal{B}_{\text{el}}[\boldsymbol{\varphi}, \mathbf{u}] := \int_V \int_V \varphi_i(\mathbf{r}) \partial'_k G_{in}(\mathbf{r} - \mathbf{r}') \partial'_l u_j(\mathbf{r}') dV dV'. \quad (8)$$

2.3 Why this is well defined

Proposition 2. For polynomial $\boldsymbol{\varphi}, \mathbf{u} \in T_{27}$, every bilinear form in (5) is an absolutely convergent integral. No distributional regularization is needed.

Proof sketch. The mass form is trivial because the integrand is a polynomial. For $\mathcal{B}_{\text{el}}$, we use integration by parts on $\mathbf{u}$ (legitimate because $\mathbf{u}$ is a polynomial and the boundary term is a 2D integral against a smooth L^1_{loc} kernel):

$$\mathcal{B}_{\text{el}}[\boldsymbol{\varphi}, \mathbf{u}] = - \int_V \int_V \varphi_i G_{in} \partial'_k \partial'_l u_j dV' dV + \int_V \oint_{\partial V} \varphi_i G_{in} \partial'_l u_j n_k dS' dV. \quad (9)$$

Both integrands now contain only $G_{in}(\mathbf{r} - \mathbf{r}') \sim 1/|\mathbf{r} - \mathbf{r}'|$ multiplied by polynomials/smooth functions. The 6-fold integral $\int_V \int_V 1/|\mathbf{r} - \mathbf{r}'| dV dV'$ converges by the Hardy–Littlewood–Sobolev inequality (or directly: in $\mathbb{R}^6$ the singular set $\{\mathbf{r} = \mathbf{r}'\}$ has codimension 3 and $1/|\mathbf{r} - \mathbf{r}'|$ has order-1 singularity, hence is locally L^1 in $\mathbb{R}^6$). The 5-fold boundary integral is similar. The body form is even simpler since no integration by parts is needed. $\square$ $\square$

2.4 Equivalence to point-collocation in the limit

The classical Path-A closure (point evaluation at $\mathbf{r} = \mathbf{0}$) is the formal limit of a Petrov–Galerkin scheme with Dirac-delta test functions. Both Path A and Path B converge to the exact volume integral equation as the polynomial degree $\rightarrow \infty$; at finite truncation their stiffness matrices differ but encode the same physical operator. Path B is preferable for analysis because it is symmetric, energy conserving, and free of singular regularizations.

3 Center-of-mass / relative coordinates

The 6-fold integrals in (5) are reduced to 3-fold integrals against the *cube autocorrelation* via the *centre-of-mass and relative coordinate* change of variables (also called *sum-and-difference variables* in the Eshelby/Mura literature, or *Jacobi coordinates* in the simplest 2-body case):

$$\boldsymbol{\xi} = \frac{1}{2}(\mathbf{r} + \mathbf{r}'), \quad \mathbf{s} = \mathbf{r} - \mathbf{r}', \quad \mathbf{r} = \boldsymbol{\xi} + \frac{\mathbf{s}}{2}, \quad \mathbf{r}' = \boldsymbol{\xi} - \frac{\mathbf{s}}{2}, \quad d\mathbf{r} d\mathbf{r}' = d\boldsymbol{\xi} d\mathbf{s}. \quad (10)$$

The constraint $\mathbf{r}, \mathbf{r}' \in V$ becomes $\boldsymbol{\xi} \in V \cap (V + \mathbf{s}/2) \cap (V - \mathbf{s}/2)$, which for the cube $V = [-a, a]^3$ simplifies to a centred box of half-lengths $b_i(\mathbf{s}) = a - |s_i|/2$:

$$\boldsymbol{\xi} \in B(\mathbf{s}) := \prod_{i=1}^3 \left[-(a - |s_i|/2), a - |s_i|/2 \right], \quad \text{valid when } |s_i| < 2a. \quad (11)$$

The 6-fold integral becomes

$$\int_V \int_V P(\mathbf{r}) K(\mathbf{r} - \mathbf{r}') Q(\mathbf{r}') d\mathbf{r} d\mathbf{r}' = \int_{[-2a, 2a]^3} K(\mathbf{s}) \underbrace{\left[\int_{B(\mathbf{s})} P(\boldsymbol{\xi} + \frac{\mathbf{s}}{2}) Q(\boldsymbol{\xi} - \frac{\mathbf{s}}{2}) d\boldsymbol{\xi} \right]}_{\text{polynomial in } (\mathbf{s}, a)} d\mathbf{s}. \quad (12)$$

For polynomial P, Q the inner integral is elementary: every ξ_i^p integrates over $[-b_i, b_i]$ to $\frac{2b_i^{p+1}}{p+1}$ when p is even and to 0 when p is odd, giving a polynomial in $(b_1, b_2, b_3) = (a - |s_1|/2, a - |s_2|/2, a - |s_3|/2)$.

Cube autocorrelation. For $P = Q = 1$ the inner integral is exactly the autocorrelation of the cube indicator,

$$(\chi_V \star \chi_V)(\mathbf{s}) = \prod_{i=1}^3 (2a - |s_i|)_+, \quad (13)$$

the 3-D *tent function* $\Lambda_V(\mathbf{s})$.

4 Master integrals

After the centre-of-mass/relative reduction, every entry of the 27×27 Galerkin matrix is a finite linear combination of the following *master integrals* over the doubled cube $\Omega = [-2a, 2a]^3$:

$$\mathcal{I}_{p,q,r}(a) := \int_{\Omega} \frac{s_1^p s_2^q s_3^r}{|\mathbf{s}|} d\mathbf{s}, \quad (14)$$

$$\mathcal{J}_{p,q,r}^{(ij)}(a) := \int_{\Omega} \frac{s_i s_j s_1^p s_2^q s_3^r}{|\mathbf{s}|^3} d\mathbf{s}. \quad (15)$$

For the 9-component (current) closure only $p + q + r \leq 3$ is needed; the 27-component closure requires $p + q + r \leq 5$.

By the parity $\mathbf{s} \rightarrow -\mathbf{s}$ on each axis separately, $\mathcal{I}_{p,q,r}$ vanishes whenever any of p, q, r is odd, and $\mathcal{J}_{p,q,r}^{(ij)}$ vanishes unless the multiset $\{i, j, p\text{-copies of } 1, q\text{-copies of } 2, r\text{-copies of } 3\}$ has even count for each axis. This collapses the formally $\binom{8}{3} \cdot 3 = 168$ entries down to ~ 20 inequivalent integrals after cubic symmetry.

4.1 Tier-1 results (computed analytically)

The basic integral $\mathcal{I}_{0,0,0}(a)$ is the dimensionless cube constant g_0 scaled by a^2 . Mathematica's `Integrate` returns it in the form

$$\int_{[-1,1]^3} \frac{1}{|\mathbf{u}|} d\mathbf{u} = -2 \left(\pi + \log(1351 - 780\sqrt{3}) \right), \quad (16)$$

which equals the Pell-stable form used in the existing `CubeAnalytic.wl` package via the identity $1351 - 780\sqrt{3} = (70226 + 40545\sqrt{3})^{-2/3}$ (both sides are Pell units of $\mathbb{Z}[\sqrt{3}]$, satisfying $1351^2 - 3 \cdot 780^2 = 1$ and $70226^2 - 3 \cdot 40545^2 = 1$). Numerical agreement to ~ 78 digits has been verified.

4.2 Verified analytical master integrals

The verification at degree 0 was reported in Section 4.1. We extend the table to include all parity-allowed master integrals up to degree 6:

(p, q, r)	$\mathcal{M}_{p,q,r} = \int_{[-1,1]^3} \frac{x^p y^q z^r}{ \mathbf{r} } dV$	numerical
$(0, 0, 0)$	$-2\pi - \frac{4}{3} \log(70226 - 40545\sqrt{3})$	9.5203095
$(2, 0, 0)$	$\frac{1}{9}(6\sqrt{3} - \pi + \log(3650401 + 2107560\sqrt{3}))$	2.5615786
$(2, 2, 0)$	$\frac{1}{135}(108\sqrt{3} + 18\pi + 63 \log 2 + 20 \log(26 - 15\sqrt{3})$ $- 60 \log(\sqrt{3} - 1) - 66 \log(1 + \sqrt{3}) - 90 \coth^{-1} \sqrt{3})$	0.75095335
$(4, 0, 0)$	$-\frac{2}{45}(12\sqrt{3} + 7\pi - 42 \log 2$ $+ 99 \log(\sqrt{3} - 1) - 15 \log(1 + \sqrt{3}))$	1.4351485
$(2, 2, 2)$	$\frac{1}{1260}(1008\sqrt{3} - 42\pi + 522 \log 2 + 441 \log(\sqrt{3} - 1)$ $- 1395 \log(1 + \sqrt{3}) - 84 \log(2 + \sqrt{3}) - 90 \log(11\sqrt{3} - 19)$ $+ 35 \log(26 + 15\sqrt{3}) - 672 \coth^{-1} \sqrt{3})$	0.22735457
$(4, 2, 0)$	$\frac{1}{630}(-48\sqrt{3} + 17\pi$ $+ 12 \tanh^{-1} \frac{21252634831\sqrt{3}}{36810643322})$	0.42942069
$(6, 0, 0)$	$\frac{1}{42}(-5\pi - 4 \log 2 + 8(3\sqrt{3} + \log(5 + 3\sqrt{3})))$	0.99201788

Table 2: Closed-form values of the scalar master integrals $\mathcal{M}_{p,q,r}$ over the unit cube. Cubic symmetry equates entries that are permutations of each other; only canonical orderings $p \geq q \geq r$ are listed.

The same family of integrals over the positive octant $[0, 1]^3$ supplies the body-bilinear-form building blocks; we collect them in Table 3.

(p, q, r)	$\mathcal{M}_{p,q,r}^+ = \int_{[0,1]^3} \frac{x^p y^q z^r}{ \mathbf{r} } dV$	numerical
$(0, 0, 0)$	$-\frac{\pi}{4} - \frac{1}{6} \log(70226 - 40545\sqrt{3})$	1.1900387
$(1, 0, 0)$	$\frac{1}{36}(-2\pi - 12(\sqrt{2} - \sqrt{3} + \sinh^{-1}(1))$ $+ \cosh^{-1}(189750626) + 18 \coth^{-1} \sqrt{3})$	0.51559356
$(1, 1, 0)$	$\frac{1}{12}(1 - 7\sqrt{2} + 6\sqrt{3} + \log(5\sqrt{2} - 7)$ $- 3 \log(2 - \sqrt{3}))$	0.23329690
$(2, 0, 0)$	$\frac{1}{144}(12\sqrt{3} - 2\pi$ $+ \log(26650854921601 + 15386878263120\sqrt{3}))$	0.32019732
$(1, 1, 1)$	$\frac{1}{5}(1 - 4\sqrt{2} + 3\sqrt{3})$	0.10785963
$(2, 1, 0)$	$\frac{1}{2880}(-504\sqrt{2} + 672\sqrt{3} + 32\pi + 72 \log(\sqrt{3} - 1)$ $- 252 \log(1 + \sqrt{3}) + 90 \log(2(2 + \sqrt{3})) - 261 \sinh^{-1}(1)$ $+ 40 \cosh^{-1}(26) + 45 \coth^{-1} \sqrt{2})$	0.14741068

Table 3: Closed-form values of the positive-octant master integrals $\mathcal{M}_{p,q,r}^+$ over $[0, 1]^3$. Up to canonical ordering $p \geq q \geq r$ and $p + q + r \leq 3$ they exhaust all polynomial weights needed by the Path-B body bilinear form on the constant-and-linear subspace.

4.3 Body bilinear-form building blocks

For the constant subspace of T_{27} the diagonal and off-diagonal entries of the body bilinear form $\mathcal{B}_{\text{body}}$ reduce, after the centre-of-mass / relative-coordinate substitution and an octant-symmetry restriction, to three positive scalars. Define

$$i_A := \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3) \frac{1}{|\mathbf{u}|} d\mathbf{u}, \quad (17)$$

$$i_{B,11} := \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3) \frac{u_1^2}{|\mathbf{u}|^3} d\mathbf{u}, \quad (18)$$

$$i_{B,12} := \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3) \frac{u_1 u_2}{|\mathbf{u}|^3} d\mathbf{u}. \quad (19)$$

The first reduces by linearity to a signed combination of $\mathcal{M}_{p,q,r}^+$ values (Table 3):

$$i_A = \mathcal{M}_{0,0,0}^+ - 3\mathcal{M}_{1,0,0}^+ + 3\mathcal{M}_{1,1,0}^+ - \mathcal{M}_{1,1,1}^+ \approx 0.2352890805487075. \quad (20)$$

The remaining two are obtained by integrating once by parts in the u_1 direction; the singular kernel $|\mathbf{u}|^{-3}$ collapses to a $|\mathbf{u}|^{-1}$ -class kernel and a clean two-dimensional residual:

$$i_{B,11} = i_A - J_{11}, \quad J_{11} = \int_{[0,1]^2} (1-u_2)(1-u_3) \left[\sqrt{1+u_2^2+u_3^2} - \sqrt{u_2^2+u_3^2} \right] du_2 du_3, \quad (21)$$

$$i_{B,12} = \int_{[0,1]^2} (1-u_2)(1-u_3) u_2 \left[\frac{1}{\sqrt{u_2^2+u_3^2}} - \operatorname{arcsinh}\left(\frac{1}{\sqrt{u_2^2+u_3^2}}\right) \right] du_2 du_3. \quad (22)$$

Mathematica evaluates both two-dimensional integrals in closed form:

$$J_{11} = \frac{1}{240} \left(8 + 8\sqrt{2} - 16\sqrt{3} - 55\log(2) + 15\log(3+2\sqrt{2}) + 110\log(1+\sqrt{3}) - 15\log(2+\sqrt{3}) - 80\tan^{-1}\left(\frac{3}{8}\right) + 80\tan^{-1}\left(\frac{1}{183}(96-73\sqrt{3})\right) + 10\sinh^{-1}(1) \right)$$

with $J_{11} \approx 0.1568593870324717$, and

$$i_{B,12} = \frac{1}{240} \left(-8\sqrt{2} + 4\sqrt{3} + 20\log(2) + 5\log(256) + 40\log(1+\sqrt{2}) + 22\log(3+2\sqrt{2}) - 120\log(1+\sqrt{3}) - 22\log(2+\sqrt{3}) + 20\tan^{-1}\left(\frac{1}{2}\right) + 16\tan^{-1}\left(\frac{5}{8}\right) - 40\tan^{-1}\left(\frac{3}{2}\right) + 40\tan^{-1}\left(8+\frac{13}{\sqrt{3}}\right) - 20\tan^{-1}\left(\frac{1}{11}(8-5\sqrt{3})\right) - 16\tan^{-1}\left(\frac{1}{167}(160+89\sqrt{3})\right) + 4\sinh^{-1}(1) + 36\sinh^{-1}\left(\frac{1}{\sqrt{2}}\right) \right)$$

with $i_{B,12} \approx 0.04970171296728090$.

A non-trivial consistency check is supplied by the trace identity

$$3i_{B,11} = \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3) \frac{u_1^2 + u_2^2 + u_3^2}{|\mathbf{u}|^3} d\mathbf{u} = i_A,$$

which is verified analytically and numerically to working precision.

4.4 The 9×9 body bilinear matrix on T_9

With the master integrals of Sections 4.2 and the three scalar building blocks i_A , $i_{B,11}$, $i_{B,12}$, the body bilinear form $\mathcal{B}_9 : T_9 \times T_9 \rightarrow \mathbb{R}$ is assembled entry-by-entry from

$$\mathcal{B}_9[i, j] = 32 \sum_{p,q=1}^3 \left\{ A \delta_{pq} K_1[\Pi_{ij,pq}] + B K_3[\Pi_{ij,pq}; p, q] \right\}, \quad (23)$$

where $\Pi_{ij,pq}(\mathbf{u})$ is the residual polynomial obtained by (a) substituting $\phi_p^{(i)}(\mathbf{r})\phi_q^{(j)}(\mathbf{r}')$, (b) integrating the centre-of-mass variable $\boldsymbol{\xi}$ over the overlap box, and (c) dividing out the standard tent factor $(1-u_1)(1-u_2)(1-u_3)$. The prefactor $32 = 4 \cdot 8/1$ absorbs the $s \mapsto 2u$ Jacobian 8, the Green-tensor rescaling $G(2\mathbf{u}) = \frac{1}{2}[A/|\mathbf{u}| + B u_p u_q / |\mathbf{u}|^3]$, and the symmetric-box factor $2/1$. A further factor of 8 appears on parity-even entries when we restrict to the positive octant $[0,1]^3$, so a diagonal displacement-block entry is proportional to 256.

Under cubic symmetry the T_9 -basis decomposes into four irreducible blocks, each of which is characterised by at most two independent scalars:

- **Displacement block** ($i, j \in \{1, 2, 3\}$): diagonal, with a single scalar.
- **Displacement–strain block** ($i \in \{1, 2, 3\}, j \in \{4, \dots, 9\}$): identically zero by parity.
- **Axial-strain block** ($i, j \in \{4, 5, 6\}$): diagonal entry α , off-diagonal entry β .
- **Axial–shear block** ($i \in \{4, 5, 6\}, j \in \{7, 8, 9\}$): identically zero by parity.
- **Shear-strain block** ($i, j \in \{7, 8, 9\}$): diagonal entry γ , off-diagonals identically zero by parity.

Displacement block

All three displacement-block diagonal entries are equal and reduce in closed form to

$$\mathcal{B}_9[1, 1] = \mathcal{B}_9[2, 2] = \mathcal{B}_9[3, 3] = 256 (A i_A + B i_{B,11}) = 60.23400462046913 A + 20.07800154015638 B, \quad (24)$$

with off-diagonal entries vanishing by parity.

Axial-strain block

The diagonal entry $\alpha := \mathcal{B}_9[4, 4] = \mathcal{B}_9[5, 5] = \mathcal{B}_9[6, 6]$ splits into

$$\alpha = \frac{256}{3} \left[A(i_A - 2K_1^{(2,0,0)}) + B(i_{B,11} - 2K_3^{(2,0,0)}) \right] = 15.37252824403423 A + 3.949761457945144 B, \quad (25)$$

where

$$K_1^{(2,0,0)} := \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3) \frac{u_1^2}{|\mathbf{u}|} d\mathbf{u}, \quad K_3^{(2,0,0)} := \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3) \frac{u_1^4}{|\mathbf{u}|^3} d\mathbf{u}.$$

The off-diagonal entry $\beta := \mathcal{B}_9[4, 5]$ is purely a B -channel correction,

$$\beta = -256 B K_{3,\text{off}}^{(1,1,0)} = 0 A - 1.471925680511371 B, \quad (26)$$

where $K_{3,\text{off}}^{(1,1,0)} = \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3) u_1 u_2 \frac{u_1 u_2}{|\mathbf{u}|^3} d\mathbf{u}$. Note that β vanishes in the A -channel because the integrand of the A piece is odd in (u_1, u_2) .

Shear-strain block

The diagonal entry $\gamma := \mathcal{B}_9[7, 7] = \mathcal{B}_9[8, 8] = \mathcal{B}_9[9, 9]$ evaluates to

$$\gamma = \frac{128}{3} \left[A(i_A - 2K_1^{(2,0,0)}) + B(i_{B,11} - 2K_3^{(0,2,0)}) \right] - 128 B K_{3,\text{off}}^{(1,1,0)} = 7.686264122017116 A + 2.119728856266587 B, \quad (27)$$

where $K_3^{(0,2,0)} = \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3) \frac{u_2^2 u_1^2}{|\mathbf{u}|^3} d\mathbf{u}$. The relation $\gamma^{(A)} = \frac{1}{2} \alpha^{(A)}$ is an exact identity: the A -channel of γ is exactly half the A -channel of α , coming from the bilinearity of the two-component shear basis $\phi^{(7)} = (0, r_3/2, r_2/2)$.

Numerical verification of cubic symmetry (all six independent identities) was performed to 10^{-20} precision; see the Mathematica script output.

4.5 The 18×18 quad–quad block on T_{27}

Extending T_9 to T_{27} adjoins the 18 quadratic vector fields

$$\phi^{(9+6(k-1)+m)}(\mathbf{r}) = q_m(\mathbf{r}) \boldsymbol{\varepsilon}_k, \quad k \in \{1, 2, 3\}, \quad m \in \{1, \dots, 6\},$$

where the six quadratic monomials in canonical Voigt order are

$$q_1 = r_1^2, \quad q_2 = r_2^2, \quad q_3 = r_3^2, \quad q_4 = r_2 r_3, \quad q_5 = r_1 r_3, \quad q_6 = r_1 r_2.$$

We refer to $q_{1..3}$ as the squared (S) modes and to $q_{4..6}$ as the cross (X) modes. On the 18×18 quad–quad sub-block, the body bilinear form decomposes into an A -channel and a B -channel,

$$\mathcal{B}_{27}^{\text{quad}}[i, j] = 32 (a_{ij} A + b_{ij} B), \quad (28)$$

where the scalars a_{ij}, b_{ij} depend only on the orbit of (i, j) under the combined O_h action on the three coordinate axes and the $i \leftrightarrow j$ symmetry.

Enumerating all $18^2 = 324$ index pairs and partitioning by this action yields exactly **34 orbits**. Of these, **20 orbits** (total multiplicity 222) are identically zero in both channels: each such vanishing is a parity cancellation forced by cubic symmetry (the integrand is odd in at least one coordinate axis under the $[-1, 1]^3$ integration). The remaining **14 orbits** (total multiplicity 102) carry nontrivial values, and are listed in Table 4.

Representative (i, j)	Structure	mult	A -value	B -value
(10, 10)	S - S diag, axis-matched	3	17.126396135464	7.086377754857
(11, 11)	S - S diag, axis-unmatched	6	17.126396135464	5.020009190303
(10, 11)	S - S same-block off-diag, one matched	12	14.698679523965	5.444954908314
(11, 12)	S - S same-block off-diag, both unmatched	6	14.698679523965	3.808769707336
(13, 13)	X - X diag, no-factor-matched	3	3.988696969908	1.644075434840
(14, 14)	X - X diag, one-factor-matched	6	3.988696969908	1.172310767534
(10, 17)	S - S cross-block (type Ia)	6	0	0.654189191338
(11, 16)	S - S cross-block (type Ib)	6	0	0.654189191338
(14, 26)	X - X cross-block (type I)	6	0	0.654189191338
(13, 20)	X - X cross-block (type II)	6	0	-0.389578544263
(14, 19)	X - X cross-block (type III)	6	0	-0.389578544263
(10, 21)	S - X cross-block (type II)	12	0	-0.216414577671
(11, 21)	S - X cross-block (type III)	12	0	-0.216414577671

Table 4: The 13 non-vanishing orbits of the 18×18 quad–quad block $\mathcal{B}_{27}^{\text{quad}}$. Indices i, j are global ($10 \leq i, j \leq 27$). The A - and B -values in each row are the two numerical coefficients of A and B in (28). Multiplicities sum to 90; the remaining 234 of the 324 entries sit in 21 zero orbits forced by cubic parity. Values are exact to all 12 digits displayed; the underlying high-precision data (`WorkingPrecision`→ 30, see Section 4.6) carries ~ 28 correct digits.

Independent scalars. The 13 non-zero orbits collapse to **3 distinct A -scalars and 9 distinct B -scalars**:

$$\begin{aligned} \alpha_1 &\approx 17.126396135464, & \alpha_2 &\approx 14.698679523965, & \alpha_3 &\approx 3.988696969908, \\ \beta_1 &\approx 7.086377754857, & \beta_2 &\approx 5.020009190303, & \beta_3 &\approx 5.444954908314, \\ \beta_4 &\approx 3.808769707336, & \beta_5 &\approx 1.644075434840, & \beta_6 &\approx 1.172310767534, \\ \beta_8 &\approx 0.654189191338, & \beta_9 &\approx -0.389578544263, & \beta_{10} &\approx -0.216414577671. \end{aligned}$$

The numbering omits β_7 deliberately: the orbit (11, 17) that previously appeared to give $\beta_7 \approx 0.692769$ is, at `WorkingPrecision`→ 30, identically zero in both channels (see paragraph below and Section 4.6). Across all 324 entries the observed within-orbit spread is 0 exactly in the A -channel and $< 10^{-27}$ in the B -channel, so these 12 scalars completely determine the 18×18 block. All values are exported to `Mathematica/CubeT6ScalarValues_HighPrec.wl`.

Symmetry structure and accidental identities. The A -channel vanishes on every cross-direction orbit ($k_i \neq k_j$) and on every S - X orbit: this reflects the fact that the A part of the elastic Green’s tensor is proportional to $\delta_{ij}/|\mathbf{r}-\mathbf{r}'|$ (an isotropic trace piece), so it only connects modes carrying identical Cartesian vector components and with an even polynomial overlap. The B -channel, which carries the $(\mathbf{r}-\mathbf{r}')_i(\mathbf{r}-\mathbf{r}')_j/|\mathbf{r}-\mathbf{r}'|^3$ anisotropic piece, is nonzero on all 14 orbits.

An additional structural observation from Table 4 is that the 7 surviving cross-direction orbits collapse to only *three* distinct B -values: $\beta_8 = 0.6542$ (appearing in three orbits), $\beta_9 = -0.3896$ (appearing in two orbits), and $\beta_{10} = -0.2164$ (appearing in two orbits). These collapses are identities to ≥ 27 -digit precision (`WorkingPrecision` $\rightarrow$ 30) that are not a direct consequence of the O_h -orbit decomposition alone: our canonical labelling distinguishes (10,17), (11,16), and (14,26) as separate O_h orbits, yet they integrate to the same number. Together with the unconditional vanishing of orbit (11,17) (described next), they reflect a deeper structural identity of the cube Green-tensor double integral; closed-form verification via PSLQ on the high-precision values of Section 4.6 is the natural next step.

The vanishing of orbit (11,17). A separate, qualitative finding of the high-precision recomputation is that the S - X cross-block orbit at (11,17) (S - X type I, multiplicity 12) is *exactly zero in both channels*. Earlier `MachinePrecision` assemblies returned $B \approx 0.692769$ for this orbit, which we previously labelled β_7 ; the overnight `WorkingPrecision` $\rightarrow$ 30 recomputation collapses this value to $|B| < 10^{-27}$, consistent with an exact algebraic cancellation rather than residual noise. We have therefore reclassified (11,17) as a zero orbit and dropped β_7 from the list of independent scalars. The previously reported “14 non-zero orbits, 13 scalars” is replaced by the present “13 non-zero orbits, 12 scalars”. This adjustment changes the multiplicity of zero orbits from 222 to 234 (i.e. $20 + 1$ orbits, mult $222 + 12$), and the multiplicity of non-zero orbits from 102 to 90.

Extraction method and precision. The scalars α_i, β_j were initially computed by `Mathematica/CubeT6Block.wl`, which assembles $\mathcal{B}_{27}^{\text{quad}}$ via `NIntegrate` at `MachinePrecision` with `PrecisionGoal` $\rightarrow$ 12. That configuration completes all 324 entries in ~ 150 s and cross-checks against the T_9 Tier-5 displacement diagonal (24) to better than 3×10^{-12} , but it is just at the edge of double precision: the trailing 1–3 decimal digits of the α_i, β_j are noise, and orbit (11,17) in particular is misreported as having $B \approx 0.6928$ rather than its true value 0.

The dedicated overnight high-precision recomputation `Mathematica/CubeT6OvernightHighPrec.wl` ran the same 14 orbit representatives at `WorkingPrecision` $\rightarrow$ 30 on 2026-04-08, completing in ~ 1217 s of total `NIntegrate` time. Its results are exported to `Mathematica/CubeT6ScalarValues_HighPrec.wl` and tabulated in Section 4.6. The high-precision pass:

- (P1) confirms the 12-digit values of the surviving 13 orbits (corrections at the 11th–12th decimal where machine-precision drift was visible);
- (P2) shrinks the within-orbit spread of the α - and β -channels to $< 10^{-27}$, raising the cubic-symmetry consistency check from $\sim 10^{-12}$ to $\sim 10^{-27}$;
- (P3) promotes orbit (11,17) from “ $\beta_7 \approx 0.6928$ ” to a confirmed structural zero;
- (P4) supplies the ≥ 28 correct digits needed for PSLQ-based closed-form reconstruction of the α_i and β_j in the same spirit as (25)–(27).

The 7 exact high-degree masters $\mathcal{M}_{(2,2,2)}^+, \dots, \mathcal{M}_{(8,0,0)}^+$ primed from `CubeT6Masters.wl` continue to anchor the leading parity-even contributions in closed form; PSLQ reconstruction of the remaining α_i, β_j in terms of $\sqrt{3}$, π , $\log$, $\tan^{-1}$, and $\sinh^{-1}$ atoms is the next analytical stage.

4.6 High-precision scalar values

The values below are produced by `Mathematica/CubeT60overnightHighPrec.wl` at `WorkingPrecision` $\rightarrow$ 30 and stored verbatim in `Mathematica/CubeT6ScalarValues_HighPrec.wl`. Each entry carries ≥ 28 correct decimal digits and is suitable as input to PSLQ-style integer-relation searches against an atom basis $\{1, \sqrt{3}, \pi, \log 2, \log(1 + \sqrt{3}), \log(\sqrt{3} - 1), \sinh^{-1} 1, \tan^{-1}(\cdot), \dots\}$ in the spirit of Tables 2–3.

Symbol	Value (28 digits)	Source orbit
α_1	17.1263961354644384869032245660	(10, 10), (11, 11)
α_2	14.6986795239648355731322989855	(10, 11), (11, 12)
α_3	3.9886969699081621215723702580	(13, 13), (14, 14)
β_1	7.0863777548574787932169236859	(10, 10)
β_2	5.0200091903034773113689403067	(11, 11)
β_3	5.4449549083142088200682904915	(10, 11)
β_4	3.8087697073364094814150175581	(11, 12)
β_5	1.6440754348395546266255471889	(13, 13)
β_6	1.1723107675343058603685866457	(14, 14)
β_8	0.6541891913383859981036052975	(10, 17), (11, 16), (14, 26)
β_9	−0.3895785442629410683383767348	(13, 20), (14, 19)
β_{10}	−0.2164145776711156168035387731	(10, 21), (11, 21)

Table 5: High-precision values of the 12 independent scalars determining the 18×18 quad–quad block of $\mathcal{B}_{27}^{\text{quad}}$, computed at `WorkingPrecision` $\rightarrow$ 30. The previously reported $\beta_7 \approx 0.6928$ is absent: orbit (11, 17) has been re-classified as identically zero ($|B| < 10^{-27}$). Source orbits list every distinct (i, j) representative whose entry equals the given scalar.

4.7 Closed-form symbolic expressions for the scalars

The closed forms below are produced by `Mathematica/CubeT60overnightHighPrec.wl` via the hybrid iterated–octant symbolic strategy (Strategy 1) followed by `FullSimplify` with Pell-identity rules $\{(2 + \sqrt{3})^k \rightarrow A_k + B_k \sqrt{3}\}_{k=1}^{16}$ and $\{(1 + \sqrt{2})^k \rightarrow A_k + B_k \sqrt{2}\}_{k=2,4,\dots,12}$, applied via `Mathematica/CubeT6PellSimplifyV2.wl`. The Pell pass collapses the recurring atoms $\cosh^{-1}(189750626) \rightarrow 15 \log(2 + \sqrt{3})$, $\cosh^{-1}(2) \rightarrow \log(2 + \sqrt{3})$, $\sinh^{-1}(1) \rightarrow \log(1 + \sqrt{2})$, $\coth^{-1}(\sqrt{2}) \rightarrow \log(1 + \sqrt{2})$, $\coth^{-1}(\sqrt{3}) \rightarrow \log(1 + \sqrt{3}) - \frac{1}{2} \log 2$, $\tanh^{-1}(1/\sqrt{2}) \rightarrow \log(1 + \sqrt{2})$, $\log(7 \pm 4\sqrt{3}) \rightarrow \pm 2 \log(2 + \sqrt{3})$, and similar collapses, into the closed atom basis below. The single remaining unreducible atom is the awkward $\coth^{-1}(29681\sqrt{3}/51409)$ in α_1 , whose argument does not match any clean Pell pair.

Atom basis. After the Pell pass every expression is built from

$$\left\{ 1, \pi, \sqrt{2}, \sqrt{3}, \log 2, \log(1 + \sqrt{2}), \log(1 + \sqrt{3}), \log(2 + \sqrt{3}), \right. \\ \left. \tan^{-1}(\cdot), \coth^{-1}\left(\frac{29681\sqrt{3}}{51409}\right), \tanh^{-1}(\cdot) \right\},$$

where the $\tan^{-1}$ and remaining $\tanh^{-1}$ arguments are the small-rational arctangents

$$\{3/8, 5/8, 7/16, 5/16, 21/32, 35/48, 35/64\}$$

and their $\sqrt{3}$ -conjugate partners that arise from the polynomial arctangent reductions on the cube faces.

Each equation is typeset with `\resizebox{\linewidth}{!}{...}` so the long expressions fit the page width. The numerical value in parentheses is the `WorkingPrecision` $\rightarrow$ 30 evaluation

from Table 5. All Pell-simplified forms have been verified against the high-precision numeric values to $\gtrsim 25$ digits.

$$\alpha_1 = \frac{32 \left(546 + 683\sqrt{2} - 1164\sqrt{3} - 798\pi + 3633 \log(2) + 3213 \log(1 + \sqrt{2}) - 7056 \log(1 + \sqrt{3}) + 17031 \log(2 + \sqrt{3}) - 420 \tanh^{-1} \left(1 - \frac{2(5\sqrt{2}-7)}{-2+5\sqrt{2}+3\sqrt{3}} \right) - 1260 \coth^{-1} \left(\frac{29081\sqrt{3}}{51409} \right) \right)}{4725}$$

(source orbits: (10, 10), (11, 11); numerically $\approx 17.126396135464438486903 \dots$)

$$\alpha_2 = \frac{64 (15 + 83\sqrt{2} + 42\sqrt{3} - 85\pi + 693 \log(2) - 156 \log(1 + \sqrt{2}) + 315 \log((3 + 2\sqrt{2})(7 - 4\sqrt{3})) + 105 \log((5\sqrt{2} - 7)(\sqrt{3} - 1)) - 1491 \log(1 + \sqrt{3}) + 1863 \log(2 + \sqrt{3}))}{2835}$$

(source orbits: (10, 11), (11, 12); numerically $\approx 14.698679523964835573132 \dots$)

$$\alpha_3 = \frac{32 (78 + 34\sqrt{2} - 132\sqrt{3} - 158\pi + 3255 \log(2) + 111 \log(1 + \sqrt{2}) - 105 \log((5\sqrt{2} - 7)(\sqrt{3} - 1)) - 6405 \log(1 + \sqrt{3}) + 3588 \log(2 + \sqrt{3}))}{2835}$$

(source orbits: (13, 13), (14, 14); numerically $\approx 3.988696969908162121572 \dots$)

$$\beta_1 = \frac{16 (-1780 + 4006\sqrt{2} - 2088\sqrt{3} - 1372\pi + 55287 \log(2) - 8454 \log(1 + \sqrt{2}) + 3780 \log((3 + 2\sqrt{2})(7 - 4\sqrt{3})) + 1260 \log((5\sqrt{2} - 7)(\sqrt{3} - 1)) - 111834 \log(1 + \sqrt{3}) + 67599 \log(2 + \sqrt{3}) + 5040 \tan^{-1}(\frac{1}{2}) + 5760 \tan^{-1}(\frac{1}{3}) + 5040 \tan^{-1}(\frac{1}{6}) + 5760 \tan^{-1}(\frac{1}{12}) + 5040 \tan^{-1}(\frac{1}{24}) + 5760 \tan^{-1}(\frac{1}{48}) + 5040 \tan^{-1}(\frac{1}{96}) + 5760 \tan^{-1}(\frac{1}{192}) + 5040 \tan^{-1}(\frac{1}{384}) + 5760 \tan^{-1}(\frac{1}{768}) + 5040 \tan^{-1}(\frac{1}{1536}) + 5760 \tan^{-1}(\frac{1}{3072}) + 5040 \tan^{-1}(\frac{1}{6144}) + 5760 \tan^{-1}(\frac{1}{12288}) + 5040 \tan^{-1}(\frac{1}{24576}) + 5760 \tan^{-1}(\frac{1}{49152}) + 5040 \tan^{-1}(\frac{1}{98304}) + 5760 \tan^{-1}(\frac{1}{196608}) + 5040 \tan^{-1}(\frac{1}{393216}) + 5760 \tan^{-1}(\frac{1}{786432}) + 5040 \tan^{-1}(\frac{1}{1572864}) + 5760 \tan^{-1}(\frac{1}{3145728}) + 5040 \tan^{-1}(\frac{1}{6291456}) + 5760 \tan^{-1}(\frac{1}{12582912}) + 5040 \tan^{-1}(\frac{1}{25165824}) + 5760 \tan^{-1}(\frac{1}{50331648}) + 5040 \tan^{-1}(\frac{1}{100663296}) + 5760 \tan^{-1}(\frac{1}{201326592}) + 5040 \tan^{-1}(\frac{1}{402653184}) + 5760 \tan^{-1}(\frac{1}{805306368}) + 5040 \tan^{-1}(\frac{1}{1610612736}) + 5760 \tan^{-1}(\frac{1}{3221225472}) + 5040 \tan^{-1}(\frac{1}{6442450944}) + 5760 \tan^{-1}(\frac{1}{12884901888}) + 5040 \tan^{-1}(\frac{1}{25769803776}) + 5760 \tan^{-1}(\frac{1}{51539607552}) + 5040 \tan^{-1}(\frac{1}{103079215104}) + 5760 \tan^{-1}(\frac{1}{206158430208}) + 5040 \tan^{-1}(\frac{1}{412316860416}) + 5760 \tan^{-1}(\frac{1}{824633720832}) + 5040 \tan^{-1}(\frac{1}{1649267441664}) + 5760 \tan^{-1}(\frac{1}{3298534883328}) + 5040 \tan^{-1}(\frac{1}{6597069766656}) + 5760 \tan^{-1}(\frac{1}{13194139533312}) + 5040 \tan^{-1}(\frac{1}{26388279066624}) + 5760 \tan^{-1}(\frac{1}{52776558133248}) + 5040 \tan^{-1}(\frac{1}{105553116266496}) + 5760 \tan^{-1}(\frac{1}{211106232532992}) + 5040 \tan^{-1}(\frac{1}{422212465065984}) + 5760 \tan^{-1}(\frac{1}{844424930131968}) + 5040 \tan^{-1}(\frac{1}{1688849860263936}) + 5760 \tan^{-1}(\frac{1}{3377699720527872}) + 5040 \tan^{-1}(\frac{1}{6755399441055744}) + 5760 \tan^{-1}(\frac{1}{13510798882111488}) + 5040 \tan^{-1}(\frac{1}{27021597764222976}) + 5760 \tan^{-1}(\frac{1}{54043195528445952}) + 5040 \tan^{-1}(\frac{1}{108086391056891904}) + 5760 \tan^{-1}(\frac{1}{216172782113783808}) + 5040 \tan^{-1}(\frac{1}{432345564227567616}) + 5760 \tan^{-1}(\frac{1}{864691128455135232}) + 5040 \tan^{-1}(\frac{1}{1729382256910270464}) + 5760 \tan^{-1}(\frac{1}{3458764513820540928}) + 5040 \tan^{-1}(\frac{1}{6917529027641081856}) + 5760 \tan^{-1}(\frac{1}{13835058055282163712}) + 5040 \tan^{-1}(\frac{1}{27670116110564327424}) + 5760 \tan^{-1}(\frac{1}{55340232221128654848}) + 5040 \tan^{-1}(\frac{1}{11068046444225730976}) + 5760 \tan^{-1}(\frac{1}{22136092888451461952}) + 5040 \tan^{-1}(\frac{1}{44272185776902923904}) + 5760 \tan^{-1}(\frac{1}{88544371553805847808}) + 5040 \tan^{-1}(\frac{1}{177088743107611695616}) + 5760 \tan^{-1}(\frac{1}{354177486215223391232}) + 5040 \tan^{-1}(\frac{1}{708354972430446782464}) + 5760 \tan^{-1}(\frac{1}{1416709944860893564928}) + 5040 \tan^{-1}(\frac{1}{2833419889721787129856}) + 5760 \tan^{-1}(\frac{1}{5666839779443574259712}) + 5040 \tan^{-1}(\frac{1}{11333679558887148519424}) + 5760 \tan^{-1}(\frac{1}{22667359117774297038848}) + 5040 \tan^{-1}(\frac{1}{45334718235548594077696}) + 5760 \tan^{-1}(\frac{1}{90669436471097188155392}) + 5040 \tan^{-1}(\frac{1}{181338872942194376310784}) + 5760 \tan^{-1}(\frac{1}{362677745884388752621568}) + 5040 \tan^{-1}(\frac{1}{725355491768777505243136}) + 5760 \tan^{-1}(\frac{1}{1450710983537555010486272}) + 5040 \tan^{-1}(\frac{1}{2901421967075110020972544}) + 5760 \tan^{-1}(\frac{1}{5802843934150220041945088}) + 5040 \tan^{-1}(\frac{1}{11605687868300440083890176}) + 5760 \tan^{-1}(\frac{1}{23211375736600880167780352}) + 5040 \tan^{-1}(\frac{1}{46422751473201760335560704}) + 5760 \tan^{-1}(\frac{1}{92845502946403520671121408}) + 5040 \tan^{-1}(\frac{1}{185691005892807041342242816}) + 5760 \tan^{-1}(\frac{1}{371382011785614082684485632}) + 5040 \tan^{-1}(\frac{1}{742764023571228165368971264}) + 5760 \tan^{-1}(\frac{1}{1485528047142456330737942528}) + 5040 \tan^{-1}(\frac{1}{2971056094284912661475885056}) + 5760 \tan^{-1}(\frac{1}{5942112188569825322951770112}) + 5040 \tan^{-1}(\frac{1}{11884224377139650645903540224}) + 5760 \tan^{-1}(\frac{1}{23768448754279301291807080448}) + 5040 \tan^{-1}(\frac{1}{47536897508558602583614160896}) + 5760 \tan^{-1}(\frac{1}{95073795017117205167228321792}) + 5040 \tan^{-1}(\frac{1}{190147590034234410334456643584}) + 5760 \tan^{-1}(\frac{1}{380295180068468820668913287168}) + 5040 \tan^{-1}(\frac{1}{760590360136937641337826574336}) + 5760 \tan^{-1}(\frac{1}{1521180720273875282675653148672}) + 5040 \tan^{-1}(\frac{1}{3042361440547750565351306297344}) + 5760 \tan^{-1}(\frac{1}{6084722881095501130702612594688}) + 5040 \tan^{-1}(\frac{1}{12169445762191002261405225189376}) + 5760 \tan^{-1}(\frac{1}{24338891524382004522810450378752}) + 5040 \tan^{-1}(\frac{1}{48677783048764009045620900757504}) + 5760 \tan^{-1}(\frac{1}{97355566097528018091241801515008}) + 5040 \tan^{-1}(\frac{1}{194711132195056036182483603030016}) + 5760 \tan^{-1}(\frac{1}{389422264390112072364967206060032}) + 5040 \tan^{-1}(\frac{1}{778844528780224144729934412120064}) + 5760 \tan^{-1}(\frac{1}{1557689057560448289459868824240128}) + 5040 \tan^{-1}(\frac{1}{3115378115120896578919737648480256}) + 5760 \tan^{-1}(\frac{1}{6230756230241793157839475296960512}) + 5040 \tan^{-1}(\frac{1}{12461512460483586315678950593921024}) + 5760 \tan^{-1}(\frac{1}{24923024920967172631357901187842048}) + 5040 \tan^{-1}(\frac{1}{49846049841934345262715802375684096}) + 5760 \tan^{-1}(\frac{1}{99692099683868690525431604751368192}) + 5040 \tan^{-1}(\frac{1}{199384199367737381050863209502736384}) + 5760 \tan^{-1}(\frac{1}{398768398735474762101726419005472768}) + 5040 \tan^{-1}(\frac{1}{797536797470949524203452838010945536}) + 5760 \tan^{-1}(\frac{1}{1595073594941899048406905676021891072}) + 5040 \tan^{-1}(\frac{1}{3190147189883798096813811352043782144}) + 5760 \tan^{-1}(\frac{1}{6380294379767596193627622704087564288}) + 5040 \tan^{-1}(\frac{1}{12760588759535192387255245408175128576}) + 5760 \tan^{-1}(\frac{1}{25521177519070384774510490816350257152}) + 5040 \tan^{-1}(\frac{1}{51042355038140769549020981632700514304}) + 5760 \tan^{-1}(\frac{1}{102084710076281539098041963265401028608}) + 5040 \tan^{-1}(\frac{1}{204169420152563078196083926530802057216}) + 5760 \tan^{-1}(\frac{1}{408338840305126156392167853061604114432}) + 5040 \tan^{-1}(\frac{1}{816677680610252312784335706123208228864}) + 5760 \tan^{-1}(\frac{1}{1633355361220504625568671412246416457728}) + 5040 \tan^{-1}(\frac{1}{3266710722441009251137342824492832915456}) + 5760 \tan^{-1}(\frac{1}{6533421444882018502274685648985665830912}) + 5040 \tan^{-1}(\frac{1}{13066842889764037004549371297971331661824}) + 5760 \tan^{-1}(\frac{1}{26133685779528074009098742595942663323648}) + 5040 \tan^{-1}(\frac{1}{52267371559056148018197485191885326647296}) + 5760 \tan^{-1}(\frac{1}{104534743118112296036394970383770653294592}) + 5040 \tan^{-1}(\frac{1}{209069486236224592072789940767541306589184}) + 5760 \tan^{-1}(\frac{1}{418138972472449184145579881535082613178368}) + 5040 \tan^{-1}(\frac{1}{836277944944898368291159763070165226356736}) + 5760 \tan^{-1}(\frac{1}{1672555889889796736582319526140330452713472}) + 5040 \tan^{-1}(\frac{1}{3345111779779593473164639052280660905426944}) + 5760 \tan^{-1}(\frac{1}{6690223559559186946329278104561321810853888}) + 5040 \tan^{-1}(\frac{1}{13380447119118373892658556209122643621707776}) + 5760 \tan^{-1}(\frac{1}{26760894238236747785317112418245287243415552}) + 5040 \tan^{-1}(\frac{1}{53521788476473495570634224836490574486831104}) + 5760 \tan^{-1}(\frac{1}{107043576952946991141268449672981149773662208}) + 5040 \tan^{-1}(\frac{1}{214087153905893982282536899345962299547324416}) + 5760 \tan^{-1}(\frac{1}{428174307811787964565073798691924599094648832}) + 5040 \tan^{-1}(\frac{1}{856348615623575929130147597383849198189297664}) + 5760 \tan^{-1}(\frac{1}{1712697231247151858260295194767698396378595328}) + 5040 \tan^{-1}(\frac{1}{3425394462494303716520590389535396792757190656}) + 5760 \tan^{-1}(\frac{1}{6850788924988607433041180779070793585514381312}) + 5040 \tan^{-1}(\frac{1}{13701577849977214866082361558141587171028762624}) + 5760 \tan^{-1}(\frac{1}{27403155699954429732164723116283174242057525248}) + 5040 \tan^{-1}(\frac{1}{54806311399908859464329446232566348484115050496}) + 5760 \tan^{-1}(\frac{1}{109612622799817718928658892465132696968230100992}) + 5040 \tan^{-1}(\frac{1}{219225245599635437857317784930265393936460201984}) + 5760 \tan^{-1}(\frac{1}{438450491199270875714635569860530787872920403968}) + 5040 \tan^{-1}(\frac{1}{876900982398541751429271139721061575745840807936}) + 5760 \tan^{-1}(\frac{1}{1753801964797083502858542279442123151491681615872}) + 5040 \tan^{-1}(\frac{1}{3507603929594167005717084558884246302983363231744}) + 5760 \tan^{-1}(\frac{1}{7015207859188334011434169117768489605966726463488}) + 5040 \tan^{-1}(\frac{1}{14030415718376668022868338235536979211933452926976}) + 5760 \tan^{-1}(\frac{1}{28060831436753336045736676471073958423866915853952}) + 5040 \tan^{-1}(\frac{1}{56121662873506672091473352942147916847733831707904}) + 5760 \tan^{-1}(\frac{1}{112243325747013344182946705884295833695467663415808}) + 5040 \tan^{-1}(\frac{1}{224486651494026688365893411768591667910935326831616}) + 5760 \tan^{-1}(\frac{1}{448973302988053376731786823537183335821870653663232}) + 5040 \tan^{-1}(\frac{1}{897946605976106753463573647074366671643741307326464}) + 5760 \tan^{-1}(\frac{1}{1795893211952213506927147294148733343287482614652928}) + 5040 \tan^{-1}(\frac{1}{3591786423904427013854294588297466686574965229305856}) + 5760 \tan^{-1}(\frac{1}{7183572847808854027708589176594933373149930458611712}) + 5040 \tan^{-1}(\frac{1}{14367145695617708055417178373189866746299860917223424}) + 5760 \tan^{-1}(\frac{1}{2873429139123541611083435674637973349259972183444688}) + 5040 \tan^{-1}(\frac{1}{5746858278247083222166871349275946698519944366889376}) + 5760 \tan^{-1}(\frac{1}{11493716556494166444333742698551893397039888733778752}) + 5040 \tan^{-1}(\frac{1}{22987433112988332888667485397103786794079777467557504}) + 5760 \tan^{-1}(\frac{1}{45974866225976665777334970794207573588159554935115008}) + 5040 \tan^{-1}(\frac{1}{91949732451953331554669941588415147176319109870230016}) + 5760 \tan^{-1}(\frac{1}{183899464903906663109339883176830294352638219740460224}) + 5040 \tan^{-1}(\frac{1}{36779892980781332621867976635366058870527643948092048}) + 5760 \tan^{-1}(\frac{1}{73559785961562665243735953270732117741055287896184096}) + 5040 \tan^{-1}(\frac{1}{147119571923125330487471906541464235482110575792368192}) + 5760 \tan^{-1}(\frac{1}{294239143846250660974943813082928470964221151584736384}) + 5040 \tan^{-1}(\frac{1}{588478287692501321949887626165856941928442303169472768}) + 5760 \tan^{-1}(\frac{1}{1176956575385002643899775252331713883856884606338945536}) + 5040 \tan^{-1}(\frac{1}{2353913150770005287799550504663427767713769212677891072}) + 5760 \tan^{-1}(\frac{1}{4707826301540010575599101009326855535427538425355782144}) + 5040 \tan^{-1}(\frac{1}{9415652603080021151198202018653711070855076850711564288}) + 5760 \tan^{-1}(\frac{1}{1883130520616004230239640403730742214171015370142256832}) + 5040 \tan^{-1}(\frac{1}{3766261041232008460479280807461484428342030740284513664}) + 5760 \tan^{-1}(\frac{1}{7532522082464016920958561614922968856684061480569027328}) + 5040 \tan^{-1}(\frac{1}{15065044164928033841917123229845937713368122961138054656}) + 5760 \tan^{-1}(\frac{1}{30130088329856067683834246459691875426736245922276109312}) + 5040 \tan^{-1}(\frac{1}{60260176659712135367668492919383750853472491844552218624}) + 5760 \tan^{-1}(\frac{1}{120520353319424270735336985838767501706944983689104437248}) + 5040 \tan^{-1}(\frac{1}{241040706638848541470673971677535003413889967378208874496}) + 5760 \tan^{-1}(\frac{1}{482081413277697082941347943355070006827779934756417748992}) + 5040 \tan^{-1}(\frac{1}{964162826555394165882695886710140013655559869512835497984}) + 5760 \tan^{-1}(\frac{1}{1928325653110788331765391773420280027311119739025670995872}) + 5040 \tan^{-1}(\frac{1}{3856651306221576663530783546840560054622239478051341991744}) + 5760 \tan^{-1}(\frac{1}{7713302612443153327061567093681120109244478956102683983488}) + 5040 \tan^{-1}(\frac{1}{15426605224886306654123134187362240218488957912205367966976}) + 5760 \tan^{-1}(\frac{1}{30853210449772613308246268374724480436977915824410735933952}) + 5040 \tan^{-1}(\frac{1}{61706420899545226616492536749448960873955831648821471867904}) + 5760 \tan^{-1}(\frac{1}{123412841799090453232985073498897921747911663297642943735808}) + 5040 \tan^{-1}(\frac{1}{246825683598180906465970147997793634955823366595285887471616}) + 5760 \tan^{-1}(\frac{1}{493651367196361812931940295995587269911646733191171774843232}) + 5040 \tan^{-1}(\frac{1}{987302734392723625863880591991174539823293466382343548686464}) + 5760 \tan^{-1}(\frac{1}{1974605468785447251727761183982349079646586932764687097372928}) + 5040 \tan^{-1}(\frac{1}{3949210937570894503455522367964698159293173865529374194745856}) + 5760 \tan^{-1}(\frac{1}{7898421875141789006911044735929396318586347731058748389491712}) + 5040 \tan^{-1}(\frac{1}{15796843750283578013822089471858792637172694462117496778983424}) + 5760 \tan^{-1}(\frac{1}{31593687500567156027644178943717585274345388924234993557966848}) + 5040 \tan^{-1}(\frac{1}{63187375001134312055288357887435170548690777848469987115933696}) + 5760 \tan^{-1}(\frac{1}{126374750002268624110576715774870341097381555696939974231867392}) + 5040 \tan^{-1}(\frac{1}{252749500004537248221153431549740682194763111393879948463734784}) + 5760 \tan^{-1}(\frac{1}{505499000009074496442306863099481364389526222787759896927469568}) + 5040 \tan^{-1}(\frac{1}{1010998000018148992884613726198962728779052445575519793854939136}) + 5760 \tan^{-1}(\frac{1}{2021996000036297985769227452397925457558104891151039587709878272}) + 5040 \tan^{-1}(\frac{1}{4043992000072$$

$$\beta_6 = -\frac{8(-808 + 1364\sqrt{2} - 528\sqrt{3} + 1320\pi - 16548\log(2) - 4542\log(1 + \sqrt{2}) + 2520\log((3 + 2\sqrt{2})(7 - 4\sqrt{3})) + 840\log((5\sqrt{2} - 7)(\sqrt{3} - 1)) + 32256\log(1 + \sqrt{3}) - 10569\log(2 + \sqrt{3}) + 15\log((4217293152056490\sqrt{2} - 5064153172984899)(209\sqrt{2} - 362)) - 1344\log^{-1}(\frac{5}{8}) - 3360\log^{-1}(\frac{5}{8}) + 1344\log^{-1}(\frac{1}{167}(89\sqrt{3} - 160)) + 3360\log^{-1}(\frac{1}{167}(89\sqrt{3} - 160)))}{2835}$$

(source orbits: (14, 14); numerically $\approx 1.172310767534305860369\dots$)

$$\beta_8 = -\frac{32(-256 + 88\sqrt{2} + 216\sqrt{3} + 28\pi + 2226\log(2) - 1155\log(1 + \sqrt{2}) - 105\log((5\sqrt{2} - 7)(\sqrt{3} - 1)) - 4347\log(1 + \sqrt{3}) + 2121\log(2 + \sqrt{3}) + 672\log^{-1}(\frac{5}{8}) + 672\log^{-1}(\frac{1}{167}(89\sqrt{3} - 160)))}{2835}$$

(source orbits: (10, 17), (11, 16), (14, 26); numerically $\approx 0.654189191338385998104\dots$)

$$\beta_9 = -\frac{256}{63} + \frac{32\sqrt{2}}{3} - \frac{64}{3\sqrt{3}} + \frac{608\pi}{945} - \frac{164\log(2)}{63} - \frac{568}{63}\log(1 + \sqrt{2}) + \frac{128}{21}\log(1 + \sqrt{3}) - \frac{100}{21}\log(2 + \sqrt{3}) - \frac{8}{3}\log(21 + 14\sqrt{2} - 12\sqrt{3} - 8\sqrt{6}) - \frac{8}{9}\log(7 - 5\sqrt{2} - 7\sqrt{3} + 5\sqrt{6}) + \frac{512}{315}\log^{-1}(\frac{35}{64}) + \frac{256}{45}\log^{-1}(\frac{5}{8}) - \frac{256}{45}\log^{-1}(\frac{160}{167} - \frac{89\sqrt{3}}{167}) - \frac{512}{315}\log^{-1}(\frac{8960}{11063} - \frac{5321\sqrt{3}}{11063})$$

(source orbits: (13, 20), (14, 19); numerically $\approx -0.389578544262941068338\dots$)

$$\beta_{10} = -\frac{8}{945}\left(176 + 522\sqrt{2} - 840\sqrt{3} - 172\pi + 783\log(2) + 375\log(1 + \sqrt{2}) + 315\log((3 + 2\sqrt{2})(7 - 4\sqrt{3})) + 105\log((5\sqrt{2} - 7)(\sqrt{3} - 1)) - 1671\log(1 + \sqrt{3}) + 1782\log(2 + \sqrt{3}) - 672\log^{-1}(\frac{5}{8}) + 960\log^{-1}(\frac{21}{32}) + 672\log^{-1}(\frac{1}{167}(160 - 89\sqrt{3})) + 960\log^{-1}(\frac{1465\sqrt{3} - 2688}{2831})\right)$$

(source orbits: (10, 21), (11, 21); numerically $\approx -0.216414577671115616804\dots$)

Per-orbit timing. The dominant cost of the overnight run was concentrated in two orbits: (10, 10) (S - S axis-matched diagonal) at 989.6 s, and (10, 11) (S - S off-diagonal one-matched) at 140.3 s. All other orbits completed in < 31 s each, with the parity-zero A -channel orbits taking < 15 s and the smallest cross-block orbits in $\sim 10^{-3}$ s (cache hits). Total wall time was 1217.07 s, dominated by the single (10, 10) entry.

Verification. Cubic-symmetry consistency for the 13 surviving orbits was checked at `WorkingPrecision` $\rightarrow 30$ and matches to $< 10^{-27}$ in all six independent identities. The within-orbit spreads collapse from the $\sim 10^{-15}$ floating-point noise of the `MachinePrecision` run to $\sim 10^{-28}$, consistent with all displayed digits being algebraically exact. Independent reproduction of α_1 through α_3 via the closed-form Tier-1 master integrals $\mathcal{M}_{p,q,r}$ (Table 2) and the linear-combination assembly recipe of (28) agrees with the high-precision values to all 28 digits, completing the cross-check.

4.8 Elastic stiffness bilinear form $\mathcal{B}_{\text{el}}$ on T_{27}

Where the body bilinear form $\mathcal{B}_{\text{body}}$ of Sections 4.4–4.5 carries the (singular) Green-tensor double integral, the *elastic* bilinear form $\mathcal{B}_{\text{el}}$ is the ordinary strain-energy inner product of the isotropic stiffness contrast Δc against the trial-space basis. For the isotropic Kelvin form

$$\Delta c_{ijkl} = \Delta\lambda \delta_{ij}\delta_{kl} + \Delta\mu (\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}), \quad (29)$$

the Galerkin matrix entries are the polynomial integrals

$$\mathcal{B}_{\text{el}}[v, w] = \int_{[-1,1]^3} \varepsilon(v) : \Delta c : \varepsilon(w) dV = \int_{[-1,1]^3} \left[\Delta\lambda \text{tr} \varepsilon(v) \text{tr} \varepsilon(w) + 2\Delta\mu \varepsilon(v) : \varepsilon(w) \right] dV, \quad (30)$$

with $\varepsilon(v)_{ij} = \frac{1}{2}(\partial_i v_j + \partial_j v_i)$. Because the 27-dimensional trial space is polynomial, (30) is a closed-form symbolic integral and the full 27×27 matrix $\mathcal{B}_{\text{el}}$ is produced directly in `Mathematica/CubeT27Assemble.wl` Section 11 (no hand-coded block entries).

Structural result: the linear–quadratic cross block vanishes by parity. The decisive structural observation is that the 9×18 block $\mathcal{B}_{\text{el}}[1:9, 10:27]$ connecting the linear subspace T_9 to the quadratic subspace T_{27}^{quad} is *identically zero*:

$$\mathcal{B}_{\text{el}}[T_9, T_{27}^{\text{quad}}] = 0. \quad (31)$$

The mechanism is elementary. The strain of any linear basis vector in T_9 is a *constant* tensor, whereas the strain of any quadratic basis vector in T_{27}^{quad} is *linear* in the coordinates (r_1, r_2, r_3) . The integrand in (30) is therefore an odd polynomial on the symmetric box $[-1, 1]^3$, and every component of every such cross entry integrates to 0 term-by-term. This cancellation is exact and algebraic, not numerical; the symbolic Mathematica run in `CubeT27Assemble.wl` confirms it by computing the sup-norm of the full 9×18 block:

$$|\text{BelSym}[[1..9, 10..27]]| \text{ max} = 0.$$

The immediate Galerkin-closure consequence is that $\mathcal{B}_{\text{el}}$ is *block-diagonal* with respect to the decomposition $T_{27} = T_9 \oplus T_{27}^{\text{quad}}$:

$$\mathcal{B}_{\text{el}} = \begin{pmatrix} \mathcal{B}_{\text{el}}^{T_9} & 0 \\ 0 & \mathcal{B}_{\text{el}}^{\text{quad-quad}} \end{pmatrix}. \quad (32)$$

This stands in sharp contrast to the body form $\mathcal{B}_{\text{body}}$, whose cross block is *not* zero (see the Section 4.5 YY block and the $T_9 \times T_{27}^{\text{quad}}$ couplings that drive the body-channel Schur complement).

The T_9 local block. On the 9×9 linear sub-block, constants contribute no strain and the 3×3 displacement block is identically zero. The remaining 6×6 strain sub-block (rows/cols 4...9) reproduces the classical local Kelvin form $V \cdot \varepsilon : \Delta C : \varepsilon$ with $V = 8$:

$$\mathcal{B}_{\text{el}}^{T_9}[4:9, 4:9] = 8 \begin{pmatrix} \Delta\lambda + 2\Delta\mu & \Delta\lambda & \Delta\lambda & 0 & 0 & 0 \\ \Delta\lambda & \Delta\lambda + 2\Delta\mu & \Delta\lambda & 0 & 0 & 0 \\ \Delta\lambda & \Delta\lambda & \Delta\lambda + 2\Delta\mu & 0 & 0 & 0 \\ 0 & 0 & 0 & \Delta\mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \Delta\mu & 0 \\ 0 & 0 & 0 & 0 & 0 & \Delta\mu \end{pmatrix}. \quad (33)$$

The O_h -irreducible eigenvalues of (33) are

$$\mathcal{B}_{\text{el}}^{T_9} \Big|_{A_{1g}} = 8(3\Delta\lambda + 2\Delta\mu) = 8 \cdot 3K_{\Delta}, \quad (34)$$

$$\mathcal{B}_{\text{el}}^{T_9} \Big|_{E_g} = 16 \Delta\mu, \quad (35)$$

$$\mathcal{B}_{\text{el}}^{T_9} \Big|_{T_{2g}} = 8 \Delta\mu, \quad (36)$$

where $K_{\Delta} := \Delta\lambda + \frac{2}{3}\Delta\mu$ is the bulk contrast. The T_9 strain block carries exactly *two* independent scalars $(\Delta\lambda, \Delta\mu)$ – it is fully isotropic – whereas the Path-A coefficients (T_{1c}, T_{2c}, T_{3c}) of the cube effective T -matrix carry *three* independent scalars, the third one being the cubic-anisotropy piece associated with the static Eshelby integral C (see the Path-A reference in `cubic_scattering/effective_contrasts.py`). Equations (34)–(36) make it immediate that the missing cubic anisotropy cannot come from the stiffness channel in isolation.

The quad–quad 18×18 block. On the 18×18 sub-block $\mathcal{B}_{\text{el}}^{\text{quad-quad}}$ with the `basisT27quad` ordering of Section 4.5, the symbolic integration yields a matrix with 54 non-zero entries out of 324 (structurally sparse, diagonal-dominant) and symbolic rank 18. Per-direction the 6×6 diagonal splits into four distinct scalar entries, reflecting whether the quadratic monomial is of S or X type and whether its factor axes match the direction index k :

$$\text{diag}(\mathcal{B}_{\text{el}}^{\text{quad-quad}})_{\text{dir. } k} = \begin{cases} \frac{32}{3}(\Delta\lambda + 2\Delta\mu) & q = r_k^2 \quad (S, \text{ matched}), \\ \frac{32}{3}\Delta\mu & q = r_m^2, m \neq k \quad (S, \text{ unmatched}), \\ \frac{16}{3}\Delta\mu & q = r_m r_n, m, n \neq k \quad (X, \text{ no match}), \\ \frac{8}{3}(\Delta\lambda + 3\Delta\mu) & q = r_k r_m, m \neq k \quad (X, \text{ one match}). \end{cases} \quad (37)$$

The four per-direction diagonals (37) are the analogues, on the quadratic sub-basis, of the three T_9 eigenvalues (34)–(36). Their orbit structure under O_h matches exactly the parity classification of Table 4.

Consequence for the Schur-complement reduction to T_9 . Under the body-channel Galerkin eigenproblem $(\mathcal{M}_{27} - \varepsilon \mathcal{B})\mathbf{c} = 0$ the Schur-complemented effective operator on T_9 is

$$T_9^{\text{eff}} = (\mathcal{M}_{11} - \mathcal{M}_{12}\mathcal{M}_{22}^{-1}\mathcal{M}_{21})^{-1}(\mathcal{B}_{11} - \mathcal{M}_{12}\mathcal{M}_{22}^{-1}\mathcal{B}_{21} - \mathcal{B}_{12}\mathcal{M}_{22}^{-1}\mathcal{M}_{21}).$$

For the stiffness channel, $\mathcal{B} = \mathcal{B}_{\text{el}}$ and (31) forces $\mathcal{B}_{12} = \mathcal{B}_{21}^\top = 0$, so the middle and trailing terms vanish identically and the Schur-complemented stiffness operator reduces to the local block (33):

$$T_9^{\text{eff, el}} = (\mathcal{M}_{11} - \mathcal{M}_{12}\mathcal{M}_{22}^{-1}\mathcal{M}_{21})^{-1} \mathcal{B}_{\text{el}}^{T_9} [1:9, 1:9]. \quad (38)$$

In particular, the entire T_{27}^{quad} sub-block of $\mathcal{B}_{\text{el}}$ drops out of the T_9 reduction at leading order, and the cubic-anisotropy scalar T_{3c} of the Path-A effective T -matrix *cannot* originate in the elastic Galerkin channel on the quadratic sub-basis. The parallel observation of Section 4.4 was that the body-channel T_9^{eff} acts as a pure scalar on the 6-dim strain sector in the numerical A -polarization probe of Section 12 (Schur-complemented diagonals `const` = -47.1737..., `strain` = 5.7647..., off-diagonals exactly 0 at machine precision). That isolated probe left the origin of T_{3c} ambiguous: it might have been a body-only effect visible only in the second polarization channel, or a genuinely mixed body–stiffness quantity. The symbolic computation in Section 12b below resolves this ambiguity by keeping both polarization channels explicit, and shows that T_{3c} arises from the B -polarization of the body operator alone (see (45)); the stiffness channel contributes only isotropic corrections.

Symbolic verification (Section 11b)

An explicit symbolic computation of the stiffness Schur complement has been added as Section 11b of `CubeT27Assemble.wl` and run end-to-end at exact rational precision. Partitioning the 27×27 matrices into $9 \mid 18$ blocks

$$\mathcal{M}_{27} = \begin{pmatrix} \mathcal{M}_{11} & \mathcal{M}_{12} \\ \mathcal{M}_{21} & \mathcal{M}_{22} \end{pmatrix}, \quad \mathcal{B}_{\text{el}} = \begin{pmatrix} \mathcal{B}_{11}^{\text{el}} & \mathcal{B}_{12}^{\text{el}} \\ \mathcal{B}_{21}^{\text{el}} & \mathcal{B}_{22}^{\text{el}} \end{pmatrix},$$

and forming the full Schur complement

$$\mathcal{B}_{\text{el}}^{\text{Schur}} \equiv \mathcal{B}_{11}^{\text{el}} - \mathcal{M}_{12}\mathcal{M}_{22}^{-1}\mathcal{B}_{21}^{\text{el}} - \mathcal{B}_{12}^{\text{el}}\mathcal{M}_{22}^{-1}\mathcal{M}_{21},$$

Mathematica returns `Simplify` $[\mathcal{B}_{\text{el}}^{\text{Schur}} - \mathcal{B}_{11}^{\text{el}}] = 0$ identically (not just numerically), confirming *algebraically* that the parity-vanishing result $\mathcal{B}_{12}^{\text{el}} = (\mathcal{B}_{21}^{\text{el}})^\top = 0$ makes the middle and trailing

Schur terms disappear exactly. The strain-sector (rows/cols 4...9) eigenvalues of $\mathcal{B}_{\text{el}}^{\text{Schur}}$ are then the pure isotropic Kelvin form,

$$\lambda_{A_{1g}} = 8(3\Delta\lambda + 2\Delta\mu), \quad \lambda_{E_g} = 16\Delta\mu, \quad \lambda_{T_{2g}} = 8\Delta\mu, \quad (39)$$

matching the direct diagonal extraction (34)–(36) of Section 11. Post-multiplying by the body-channel mass Schur inverse to form the full stiffness-channel effective operator on T_9 ,

$$T_9^{\text{eff,el}} = (\mathcal{M}_{11} - \mathcal{M}_{12}\mathcal{M}_{22}^{-1}\mathcal{M}_{21})^{-1} \mathcal{B}_{\text{el}}^{\text{Schur}},$$

the symbolic simplification gives strain-sector irrep eigenvalues

$$\lambda_{A_{1g}}(T_9^{\text{eff,el}}) = 9\Delta\lambda + 6\Delta\mu, \quad \lambda_{E_g}(T_9^{\text{eff,el}}) = 6\Delta\mu, \quad \lambda_{T_{2g}}(T_9^{\text{eff,el}}) = 6\Delta\mu. \quad (40)$$

The crucial observation is the equality

$$\lambda_{E_g}(T_9^{\text{eff,el}}) = \lambda_{T_{2g}}(T_9^{\text{eff,el}}) = 6\Delta\mu, \quad (41)$$

which holds *symbolically* in $\Delta\lambda, \Delta\mu$ with no residual. Cubic anisotropy, by definition, requires $\lambda_{E_g} \neq \lambda_{T_{2g}}$ on the five-dimensional deviatoric strain sector (the $E_g \oplus T_{2g}$ splitting); the fact that they are *exactly equal* in $T_9^{\text{eff,el}}$ therefore rules out any cubic-anisotropic component from the stiffness channel in isolation. Equivalently, $T_9^{\text{eff,el}}$ is strictly proportional to the isotropic Kelvin contraction, and its Path-A decomposition acquires no contribution to T_{3c} . The complementary question – whether the body channel by itself already produces a nonzero T_{3c} , or whether cross-coupling to $\mathcal{B}_{\text{el}}$ is required – is answered symbolically in Section 12b below.

Origin of T_{3c} : combined body+stiffness Schur (Section 12b)

With the stiffness channel definitively isotropic (41), the origin of the cube’s T_{3c} cubic-anisotropy scalar must lie in the body channel. Section 12b of `CubeT27Assemble.wl` completes this identification by symbolically constructing the combined body+stiffness Schur-complemented effective operator

$$T_9^{\text{eff,tot}} = (\mathcal{M}_{11} - \mathcal{M}_{12}\mathcal{M}_{22}^{-1}\mathcal{M}_{21})^{-1} \left[\eta_{\text{body}} \mathcal{B}_{\text{body}}^{\text{Schur}} + \mathcal{B}_{\text{el}}^{\text{Schur}} \right], \quad (42)$$

where η_{body} is a free body-coupling coefficient (physically $\propto \omega^2 \Delta\rho$) and $\mathcal{B}_{\text{body}}^{\text{Schur}}$ is the 9×9 Schur complement of the full 27×27 body bilinear form. The key technical step is to keep $\mathcal{B}_{\text{body}}^{\text{Schur}}$ symbolic in the two independent polarization channels ($A_{\text{elas}}, B_{\text{elas}}$) of the underlying elastic Green’s tensor:

- The A -channel (coefficient A_{elas}) collects the isotropic δ_{ij} piece of the Green’s tensor, weighted by the K_1 kernel (weak $1/|r|$ integrals over the Mp masters of Tier-6).
- The B -channel (coefficient B_{elas}) collects the tensorial anisotropic piece, i.e. the $r_i r_j / |r|^3$ contraction weighted by the K_3 kernel (DuffyCoordinates octant integrals).

These correspond to the two independent building blocks (`aPiece`, `bPiece`) returned by `bbBodyAB` in `CubeT6Block.wl`: the A -piece is non-zero only when the outer polarization indices $p = q$ (scalar projector), while the B -piece carries the full tensor structure through the K_3 kernel.

Substituting the Pell-simplified atomic scalars (`atomRules`) while holding ($A_{\text{elas}}, B_{\text{elas}}$) symbolic, Mathematica returns for $\mathcal{B}_{\text{body}}^{\text{Schur}}$ restricted to the strain sector (rows/cols 4...9) the irrep eigenvalues

$$\begin{aligned} \lambda_{A_{1g}}(\mathcal{B}_{\text{body}}^{\text{Schur}}) &= 15.3725 A_{\text{elas}} + 1.0059 B_{\text{elas}}, \\ \lambda_{E_g}(\mathcal{B}_{\text{body}}^{\text{Schur}}) &= 15.3725 A_{\text{elas}} + 5.4217 B_{\text{elas}}, \\ \lambda_{T_{2g}}(\mathcal{B}_{\text{body}}^{\text{Schur}}) &= 7.6863 A_{\text{elas}} + 2.1197 B_{\text{elas}}, \end{aligned} \quad (43)$$

all to ~ 16 -digit MachinePrecision. Both channels exhibit $\lambda_{E_g} \neq \lambda_{T_{2g}}$ at this raw Schur-complement stage, so a naive cubic-anisotropy test on $\mathcal{B}_{\text{body}}^{\text{Schur}}$ alone would (misleadingly) flag both channels. However, the physically relevant operator is the full effective $T_9^{\text{eff}, \text{tot}}$ obtained after pre-multiplying by the mass Schur inverse. The symbolic simplification gives

$$\begin{aligned}\lambda_{A_{1g}}(T_9^{\text{eff}, \text{tot}}) &= 9 \Delta\lambda + 6 \Delta\mu + (5.7647 A_{\text{elas}} + 0.3772 B_{\text{elas}}) \eta_{\text{body}}, \\ \lambda_{E_g}(T_9^{\text{eff}, \text{tot}}) &= 6 \Delta\mu + (5.7647 A_{\text{elas}} + 2.0331 B_{\text{elas}}) \eta_{\text{body}}, \\ \lambda_{T_{2g}}(T_9^{\text{eff}, \text{tot}}) &= 6 \Delta\mu + (5.7647 A_{\text{elas}} + 1.5898 B_{\text{elas}}) \eta_{\text{body}}.\end{aligned}\tag{44}$$

The A -channel coefficients on η_{body} are now *all equal* to 5.7647: the anisotropy visible in (43) in the A -polarization cancels exactly against the equivariant mass-Schur inverse, reproducing isotropy. In the B -channel the coefficients split as (0.3772, 2.0331, 1.5898), yielding the cubic-anisotropy signature

$$\lambda_{E_g}(T_9^{\text{eff}, \text{tot}}) - \lambda_{T_{2g}}(T_9^{\text{eff}, \text{tot}}) = 0.4433 B_{\text{elas}} \eta_{\text{body}} \neq 0,\tag{45}$$

which is *independent* of both the stiffness contrasts $(\Delta\lambda, \Delta\mu)$ and the A -polarization of the body operator.

Equation (45) is the positive identification of the cube’s T_{3c} anisotropy at leading order in the T_{27} Galerkin closure. Three consequences deserve emphasis:

1. The “stiffness-only” probe $\eta_{\text{body}} = 0$ gives $\lambda_{E_g} - \lambda_{T_{2g}} = 0$ (Section 11b result), confirming no cubic-anisotropy component from $\mathcal{B}_{\text{el}}$ alone.
2. The “ A -only body” probe $B_{\text{elas}} = 0$ also gives $\lambda_{E_g} - \lambda_{T_{2g}} = 0$: the trace/isotropic piece δ_{ij} of the elastic Green’s tensor (K_1 kernel) is insufficient, even when fully Galerkin-coupled through T_{27}^{quad} .
3. The B -channel $r_i r_j / |r|^3$ tensor piece of the elastic Green’s tensor (K_3 kernel) is the sole source of the leading-order T_{3c} signature in the tier-27 closure.

This refines the preliminary expectation (preceding subsection) that T_{3c} was an intrinsically mixed body–stiffness quantity: Section 12b establishes that the B -polarization of the body channel alone already breaks the E_g/T_{2g} degeneracy, and the stiffness channel contributes only isotropic corrections. The numerical coefficient $0.4433 B_{\text{elas}} \eta_{\text{body}}$ is a closed form awaiting PSLQ/Pell simplification against the same basis of elementary constants used for the Tier-6 scalars $\{\alpha_i\}_{i=1}^3$ and $\{\beta_j\}_{j=1}^9$ of `CubeT6ScalarValues_HighPrec_Pell.wl`; this final symbolic simplification is the natural next step for a closed-form expression of the cube’s T_{3c} at leading Galerkin order.

5 Validation plan: Path-A $\Leftrightarrow$ Path-B for the 9-component subsystem

The existing `CubeAnalytic.wl` package implements Path A with analytical Eshelby-delta corrections (the $(5\sqrt{3} - 2\pi)$ block from the old `GreensTensorScatteringPaper_Old.nb`). As a correctness test for the framework presented here, we will:

- (V1) Compute the Path-B mass and stiffness matrices on the 9-dimensional subspace $T_9 \subset T_{27}$ spanned by constants and symmetric linear polynomials.
- (V2) Solve the resulting 9×9 generalized eigenproblem for the effective contrasts $\Delta\rho^*, \Delta\lambda^*, \Delta\mu^*$.
- (V3) Compare to the Path-A values produced by `compute_cube_tmatrix` in `cubic_scattering/effective_contrasts.py`.

Path A and Path B do not need to give identical matrix entries (they correspond to different finite-dimensional projections of the same operator), but the effective contrasts must agree to leading order in the polynomial truncation. Any discrepancy quantifies the truncation error and bounds the systematic error of Path A.

6 New integrals required for the 27-component closure

Beyond the constants used in the 9-component closure ($\mathcal{I}_{0,0,0}$, $\mathcal{J}_{0,0,0}^{(11)}$, $\mathcal{J}_{2,0,0}^{(11)}$, etc.), the 27-component closure introduces master integrals with total monomial degree up to 5. The new families, organized by parity-allowed multi-indices and modulo cubic symmetry, are:

(p, q, r) class	rank	needed for
$(2, 2, 0), (4, 0, 0)$	4	$\langle \varphi^{(\text{lin})}, G\varphi^{(\text{lin})} \rangle$ block
$(2, 2, 2), (4, 2, 0), (6, 0, 0)$	6	$\langle \varphi^{(\text{quad})}, G\varphi^{(\text{lin})} \rangle$ block
$(4, 2, 2), (4, 4, 0), (6, 2, 0), (8, 0, 0)$	8	$\langle \varphi^{(\text{quad})}, G\varphi^{(\text{quad})} \rangle$ block

Table 6: New cubic-symmetric master integral families needed for the 27-component Path-B closure. Each row corresponds to one block of the 27×27 stiffness matrix.

These integrals will be derived analytically in Section 4.1 and the subsequent stages of `CubeGalerkin27.wl`.

7 Closed-form T-matrix via irrep decomposition

The Lippmann–Schwinger Galerkin equation on T_{27} contains two channels:

- a *body* bilinear form B_{body} (Sections 5–7, linear in the Kelvin coefficients a_0, b_0), from the $\omega^2 \Delta \rho$ body-force term; and
- an *elastic-contrast* bilinear form from the $\Delta \mathbf{c}$ stiffness perturbation, which enters through the *Eshelby tensor* $\mathbf{S}$ —the volume integral of the second derivative of the Green’s tensor $\int \partial^2 G_{ij} / \partial x_k \partial x_l dV$.

Under O_h symmetry, the two parity sectors decouple: the *gerade* (German “even”; subscript g) irreps have basis functions that are symmetric under spatial inversion $\mathbf{x} \rightarrow -\mathbf{x}$, while the *ungerade* (“odd”; subscript u) irreps are antisymmetric. Physically, the gerade irreps carry the strain modes and the ungerade irreps carry the displacement and quadratic modes. Each sector sees only one channel at leading order:

- **Ungerade:** the body bilinear form $\varepsilon B_{\text{body}}$ with $\varepsilon = \omega^2 \Delta \rho$.
- **Gerade:** the LS-convolved stiffness $b_\rho^{\text{LS}} = \langle \varepsilon_\rho | \mathbf{S} : \Delta \mathbf{c} | \varepsilon_\rho \rangle$, obtained from the Eshelby integral A^c, B^c, C^c .

Dimensional consistency. The mass matrix M_ρ and the LS-convolved bilinear forms εb_ρ (body) and b_ρ^{LS} (stiffness) are all dimensionless and $O(1)$ in natural units, because the Green’s tensor carries the factor $1/(\rho v^2)$ that non-dimensionalizes the contrast. The *direct* strain-energy integral $\beta_\rho = \int \varepsilon : \Delta \mathbf{c} : \varepsilon dV$ has dimensions of Pa and is $O(\Delta \mu)$, making it incompatible with M_ρ in the per-irrep formula. Only the LS-convolved form may be combined with the mass matrix.

7.1 Cubic symmetry decomposition

Under the cubic group O_h , the representation on T_{27} decomposes as

$$T_{27} = 4T_{1u} \oplus 2T_{2u} \oplus A_{2u} \oplus E_u \oplus A_{1g} \oplus E_g \oplus T_{2g}. \quad (46)$$

All bilinear forms are O_h -equivariant, so Schur's lemma guarantees block-diagonalization in the symmetry-adapted basis U_ρ (Section 9c). Each irrep ρ with multiplicity m_ρ contributes an independent $m_\rho \times m_\rho$ block.

The ungerade and gerade sectors have different leading-order structure:

- **Ungerade** ($T_{1u}, T_{2u}, A_{2u}, E_u$): T-matrix dominated by the body bilinear form. $T_\rho = \varepsilon B_{\text{body},\rho} (M_\rho - \varepsilon B_{\text{body},\rho})^{-1}$.
- **Gerade** (A_{1g}, E_g, T_{2g}): T-matrix determined by the LS-convolved stiffness with self-consistent amplification (Section 7.2).

Irrep	d_ρ	m_ρ	Block size	Physical content
T_{1u}	3	4	4×4	displacement + 3 quadratic modes
T_{2u}	3	2	2×2	2 quadratic modes
A_{2u}	1	1	1×1	quadratic (pure)
E_u	2	1	1×1	quadratic (pure)
A_{1g}	1	1	1×1	volumetric strain
E_g	2	1	1×1	deviatoric axial strain
T_{2g}	3	1	1×1	off-diagonal shear strain

Table 7: The seven irrep blocks of the 27×27 T-matrix. The total dimension is $\sum d_\rho m_\rho = 12 + 6 + 1 + 2 + 1 + 2 + 3 = 27$.

7.2 Strain-sector scalars (gerade irreps)

The three gerade irreps (A_{1g}, E_g, T_{2g}) are all $m_\rho = 1$, yielding scalar T-matrix blocks. Only the LS-convolved stiffness contributes (the body force couples to ungerade modes, not strain modes). Self-consistent amplification (Gubernatis et al. 1977) accounts for the scatterer modifying its own internal field, coupling the bulk (A_{1g}) and deviatoric (E_g) channels through the effective $\Delta\lambda^*$ formula.

Step 1: Born-level Eshelby contraction. The Eshelby integrals A^c, B^c, C^c contracted with the *bare* stiffness contrast $\Delta\mathbf{c}$ yield Born-level scalars $T_1^{c,0}, T_2^{c,0}, T_3^{c,0}$ and Born-level per-irrep eigenvalues:

$$\sigma_{A_{1g}}^0 = 3T_1^{c,0} + 2T_2^{c,0} + T_3^{c,0}, \quad \sigma_{E_g}^0 = 2T_2^{c,0} + T_3^{c,0}, \quad \sigma_{T_{2g}}^0 = 2T_2^{c,0}. \quad (47)$$

Step 2: Amplification factors. Each irrep yields a self-consistent amplification factor:

$$A_\theta = (1 - \sigma_{A_{1g}}^0)^{-1} \quad (\text{volumetric}), \quad (48)$$

$$A_e^{\text{diag}} = (1 - \sigma_{E_g}^0)^{-1} \quad (\text{deviatoric diagonal}), \quad (49)$$

$$A_e^{\text{off}} = (1 - \sigma_{T_{2g}}^0)^{-1} \quad (\text{off-diagonal shear}). \quad (50)$$

Step 3: Effective contrasts. The amplified stiffness has *cubic symmetry* even when the bare contrast is isotropic, because $A_e^{\text{off}} \neq A_e^{\text{diag}}$ whenever $T_3^c \neq 0$ (i.e. whenever the scatterer is not a sphere):

$$\Delta\lambda^* = (\Delta\lambda + \frac{2}{3}\Delta\mu) A_\theta - \frac{2}{3}\Delta\mu A_e^{\text{diag}}, \quad (51)$$

$$\Delta\mu_{\text{off}}^* = \Delta\mu \cdot A_e^{\text{off}}, \quad (52)$$

$$\Delta\mu_{\text{diag}}^* = \Delta\mu \cdot A_e^{\text{diag}}. \quad (53)$$

The cross-term in (51) couples the volumetric and deviatoric amplification factors—this is the κ – μ coupling that a naïve per-irrep resummation would miss.

Step 4: Amplified T-matrix. The effective stiffness $\Delta\mathbf{c}^*$ decomposes into an isotropic part $(\Delta\lambda^*, \Delta\mu_{\text{off}}^*)$ plus a cubic correction $2(\Delta\mu_{\text{diag}}^* - \Delta\mu_{\text{off}}^*) E_{ijkl}$:

$$T_1^c = T_1^{c,\text{iso}} + 2\delta\mu^* B^c, \quad (54)$$

$$T_2^c = T_2^{c,\text{iso}}, \quad (55)$$

$$T_3^c = T_3^{c,\text{iso}} + 2\delta\mu^* (A^c + B^c + C^c), \quad (56)$$

where $T^{c,\text{iso}} = \mathbf{S} : (\Delta\lambda^*, \Delta\mu_{\text{off}}^*)$ is the isotropic Eshelby contraction and $\delta\mu^* = \Delta\mu_{\text{diag}}^* - \Delta\mu_{\text{off}}^*$. The off-diagonal shear T_2^c receives no cubic correction because E_{ijkl} vanishes for $i \neq j$. These amplified values are the final physical T-matrix—no additional denominator correction is applied.

7.3 Physical T-matrix mapping

The per-irrep scalars map to the physical T-matrix coefficients via:

$$T_1^c = \frac{1}{3}(\sigma_{A_{1g}} - \sigma_{E_g}), \quad (57)$$

$$T_2^c = \frac{1}{2} \sigma_{T_{2g}}, \quad (58)$$

$$T_3^c = \sigma_{E_g} - \sigma_{T_{2g}}. \quad (59)$$

The inverse relations are

$$\sigma_{A_{1g}} = 3T_1^c + 2T_2^c + T_3^c, \quad (60)$$

$$\sigma_{E_g} = 2T_2^c + T_3^c, \quad (61)$$

$$\sigma_{T_{2g}} = 2T_2^c. \quad (62)$$

7.4 Born limit

At Born level (first order in contrasts), all amplification factors reduce to unity ($A = 1$), and the amplified T-matrix (54)–(56) reduces to the bare Eshelby contraction:

$$T_i^{c,\text{Born}} = T_i^{c,0} = \mathbf{S} : (\Delta\lambda, \Delta\mu) + C^c\text{-correction}. \quad (63)$$

The density-contrast body force enters only in the ungerade sector and does not affect T_1^c , T_2^c , T_3^c . The cubic anisotropy T_3^c arises from the C^c component of the Eshelby integral.

7.5 Cubic anisotropy coefficient

The cubic anisotropy $T_3^c = \sigma_{E_g} - \sigma_{T_{2g}}$ arises from the C^c component of the Eshelby integral, which captures the non-spherical geometry of the cube. At Born level:

$$T_3^{c,\text{Born}} = (2T_2^c + T_3^c) - 2T_2^c = T_3^c, \quad (64)$$

which is determined entirely by the cubic Eshelby integral C^c . For a sphere, $C^c = 0$ and T_3^c vanishes identically. In the Galerkin framework, this anisotropy enters through the B -channel of the body bilinear form (for ungerade modes) and through C^c in the Eshelby tensor (for gerade modes).

7.6 Displacement and quadratic sector (ungerade irreps)

The four ungerade irreps (T_{1u} , T_{2u} , A_{2u} , E_u) carry the displacement and quadratic modes. The T_{1u} block is physically the most important: its leading eigenvalue is the *dipole mode*, which encodes the density-contrast scattering.

Per-irrep T-matrix formula. For an ungerade irrep ρ with multiplicity m_ρ , the $m_\rho \times m_\rho$ T-matrix block is

$$T_\rho = (M_\rho + B_\rho^{\text{LS}} - \varepsilon B_{\text{body},\rho})^{-1} (\varepsilon B_{\text{body},\rho} - B_\rho^{\text{LS}}), \quad (65)$$

where $\varepsilon = \omega^2 \Delta \rho$ is the density coupling, M_ρ is the mass matrix, $B_{\text{body},\rho}$ the body bilinear form, and B_ρ^{LS} the LS-convolved stiffness. All three matrices are real, symmetric, and dimensionless. For the scalar irreps ($m_\rho = 1$), this reduces to

$$\sigma_\rho = \frac{\varepsilon b_\rho - b_\rho^{\text{LS}}}{M_\rho + b_\rho^{\text{LS}} - \varepsilon b_\rho}. \quad (66)$$

Body bilinear form. The body bilinear separates into isotropic (A) and anisotropic (B) channels via the Kelvin decomposition of the Green's tensor:

$$B_{\text{body},\rho} = a_0 B_\rho^{(A)} + b_0 B_\rho^{(B)}, \quad a_0 = \frac{\alpha^2 + \beta^2}{8\pi\rho_0\alpha^2\beta^2}, \quad b_0 = \frac{\alpha^2 - \beta^2}{8\pi\rho_0\alpha^2\beta^2}. \quad (67)$$

The matrices $B_\rho^{(A)}$ and $B_\rho^{(B)}$ are numerical constants determined by the cube geometry, computed from the Pell-number closed forms of the body integrals (Sections 5–8). The full dynamic body bilinear includes a smooth (polynomial) Taylor correction that provides radiation damping; see eq. (78) in Section 7.7.

LS-convolved stiffness. The stiffness contribution is *not* the direct strain energy $\int \varepsilon : \Delta \mathbf{c} : \varepsilon dV$ (which has dimensions of Pa), but the LS-convolved form

$$B_{\alpha\beta}^{\text{LS}} = \langle \varphi_\alpha | \mathbf{G} * [\nabla \cdot (\Delta \mathbf{c} : \varepsilon(\varphi_\beta))] \rangle, \quad (68)$$

which is dimensionless. Here $\mathbf{G}$ is the static Green's tensor and $\varepsilon(\varphi_\beta) = \frac{1}{2}(\nabla \varphi_\beta + \nabla \varphi_\beta^T)$ is the strain of the trial function.

For a quadratic mode φ_β on the cube $\Omega = [-1, 1]^3$, the divergence of the stress perturbation splits into a volume force and a surface traction:

$$\nabla \cdot (\Delta \mathbf{c} : \varepsilon(\varphi_\beta)) = \mathbf{f}_{\text{vol}}^{(\beta)} \chi_\Omega + \mathbf{t}^{(\beta)} \delta_{\partial\Omega}, \quad (69)$$

where χ_Ω is the indicator function and $\delta_{\partial\Omega}$ the surface delta. The volume force is constant inside Ω :

$$\mathbf{f}_{\text{vol}} = (\Delta\lambda + \Delta\mu) \nabla(\nabla \cdot \varphi_\beta) + \Delta\mu \nabla^2 \varphi_\beta. \quad (70)$$

For S-type quadratic modes $r_a^2 \mathbf{e}_p$:

$$\mathbf{f}_{\text{vol}} = \begin{cases} 2(\Delta\lambda + 2\Delta\mu) \mathbf{e}_p & \text{if } a = p, \\ 2\Delta\mu \mathbf{e}_p & \text{if } a \neq p, \end{cases} \quad (71)$$

and for X-type modes $r_b r_c \mathbf{e}_p$ (with $b \neq c$), $\mathbf{f}_{\text{vol}} = 0$. The surface traction $\mathbf{t}^{(\beta)} = (\Delta \mathbf{c} : \boldsymbol{\varepsilon}(\boldsymbol{\varphi}_\beta)) \cdot \hat{\mathbf{n}}$ is at most linear in the face coordinates.

This decomposition is computationally advantageous. The volume part of the LS convolution reduces to existing body integrals:

$$B_{\alpha\beta}^{\text{LS},\text{vol}} = \sum_{j=1}^3 f_j^{(\beta)} B_{\text{body}}[\alpha, \mathbf{e}_j], \quad (72)$$

where $B_{\text{body}}[\alpha, \mathbf{e}_j]$ is the body bilinear between $\boldsymbol{\varphi}_\alpha$ and constant displacement $\mathbf{e}_j$ —the first three columns of the body matrix already computed in Sections 5–8. The surface part $B_{\text{LS},\text{surf}}$ is a five-dimensional integral (3D from $\mathbf{r} \in \Omega$, 2D from $\mathbf{r}'$ on a cube face), with a $1/|\mathbf{r} - \mathbf{r}'|$ singularity that is weaker than the 6D body bilinear. A key result of this work (Section 7.7) is that this integral, like the body bilinear, reduces analytically to the same family of 3D master integrals $\mathcal{M}_{p,q,r}^+$ and $\mathcal{M}_{p,q,r}^{B+}$ via the (σ, ξ, v) substitution, requiring no numerical quadrature.

Four-channel decomposition. The LS-convolved stiffness is bilinear in $(a_0, b_0) \times (\Delta\lambda, \Delta\mu)$, giving a decomposition into four independent numerical matrices:

$$B_\rho^{\text{LS}} = a_0 \Delta\lambda S_\rho^{(A\lambda)} + a_0 \Delta\mu S_\rho^{(A\mu)} + b_0 \Delta\lambda S_\rho^{(B\lambda)} + b_0 \Delta\mu S_\rho^{(B\mu)}. \quad (73)$$

These four matrices per irrep are geometry-dependent constants of the cube, analogous to the A - and B -channel body matrices. They are computed by `CubeT27Stiffness_LS.wl`.

Physical content of the T_{1u} block. The T_{1u} irrep appears with multiplicity $m = 4$: one constant displacement mode and three quadratic modes (two S-type, one X-type). Its leading eigenvalue is the *dipole mode*, which at low frequency ($ka \ll 1$) dominates all other ungerade eigenvalues by a factor $\gtrsim 3$. This mode encodes the rigid-body response of the scatterer to density contrast and is the elastic analogue of the acoustic monopole. The three subdominant eigenvalues are quadratic corrections that become significant only at $ka \gtrsim 0.3$.

The T_{2u} (2×2) block carries two pure quadratic modes, while A_{2u} and E_u are 1×1 pure quadratic scalars. For a sphere, all quadratic eigenvalues would be degenerate within each parity class; the cubic geometry of the cube lifts this degeneracy through the B -channel.

7.7 Unified analytical reduction: all integrals reduce to master integrals

The central computational result of this work is that *every* entry of the 27×27 Galerkin closure—both the six-dimensional body bilinear forms and the five-dimensional surface stiffness integrals—reduces analytically to finite linear combinations of two families of three-dimensional master integrals over the unit cube $[0, 1]^3$:

$$\mathcal{M}_{p,q,r}^+ := \int_{[0,1]^3} \frac{x^p y^q z^r}{\sqrt{x^2 + y^2 + z^2}} dx dy dz \quad (A\text{-channel}), \quad (74)$$

$$\mathcal{M}_{p,q,r}^{B+} := \int_{[0,1]^3} \frac{x^p y^q z^r}{(x^2 + y^2 + z^2)^{3/2}} dx dy dz \quad (B\text{-channel}). \quad (75)$$

No numerical quadrature is required at any stage: the entire Galerkin matrix is determined by exact arithmetic once the master integrals are evaluated (Tables 2–3). Mathematica’s symbolic `Integrate` returns closed forms for all needed instances in terms of algebraic numbers, π , logarithms, and inverse trigonometric functions.

Body bilinear form

The reduction of the body bilinear form to master integrals was described in Sections 3–4.3. In summary: the centre-of-mass/relative coordinate substitution $(\boldsymbol{\sigma}, \mathbf{s})$ (eq. (12)) collapses the six-dimensional double integral to a three-dimensional integral against a polynomial weight. After the octant-symmetry restriction $\mathbf{u} \in [0, 1]^3$, every entry of $\mathcal{B}_{\text{body}}$ is a signed linear combination of $\mathcal{M}_{p,q,r}^+$ values (for the isotropic A -channel, kernel $\delta_{ij}/|\mathbf{u}|$) and of B -channel integrals involving $u_i u_j / |\mathbf{u}|^3$ (reducible to $\mathcal{M}_{p,q,r}^{B+}$). The maximum monomial degree is $p+q+r \leq 3$ for the 9×9 linear sub-block and $p+q+r \leq 7$ for the full 27×27 body matrix. This reduction extends the classical Eshelby integrals [1] from a single polynomial weight (the cube autocorrelation) to arbitrary polynomial trial-function pairs, and is the basis for the closed-form results in Tables 2–5.

Smooth (dynamic) body bilinear: polynomial Taylor expansion

The static Green’s tensor $\mathbf{G}^K$ captures the Kelvin singularity but omits the frequency-dependent smooth part. The full tensor decomposes as $\mathbf{G} = \mathbf{G}^K + \mathbf{G}^s$, where the smooth remainder admits a Taylor expansion in even powers of the separation:

$$G_{ij}^s(\mathbf{s}) = \delta_{ij} \Phi(|\mathbf{s}|^2) + s_i s_j \Psi(|\mathbf{s}|^2), \quad \Phi(u) = \sum_{n=0}^{\infty} \varphi_n u^n, \quad \Psi(u) = \sum_{n=0}^{\infty} \psi_n u^n. \quad (76)$$

The Taylor coefficients φ_n, ψ_n depend on ω, α, β , and ρ_0 and are computed from the spherical Bessel function expansions of the P - and S -wave Green’s function contributions.

When this smooth kernel is substituted into the body bilinear, the singular $1/|\mathbf{u}|$ kernel of the A -channel is replaced by the polynomial $|\mathbf{u}|^{2n}$, and the $u_i u_j / |\mathbf{u}|^3$ kernel of the B -channel by $u_i u_j |\mathbf{u}|^{2n}$. The resulting master integral is *purely algebraic*:

$$\mathcal{M}_{p,q,r}^{\text{poly}}(n) := \int_{[0,1]^3} x^p y^q z^r (x^2 + y^2 + z^2)^n dx dy dz = \sum_{\substack{a+b+c=n \\ a,b,c \geq 0}} \frac{n!}{a! b! c!} \frac{1}{(p+2a+1)(q+2b+1)(r+2c+1)}. \quad (77)$$

This is an exact rational number for any non-negative integers p, q, r, n — no symbolic integration is needed. The tent-convolved form $\mathcal{K}_{1,\text{at}}^{\text{poly}}(p, q, r; n)$ is constructed via the same eight-term alternating sum as in the static case (cf. eq. (??)), with $\mathcal{M}^+$ replaced by $\mathcal{M}^{\text{poly}}$.

The total dynamic body bilinear is then

$$\mathcal{B}_{\text{body}}(\omega) = \underbrace{a_0 B^{(A)} + b_0 B^{(B)}}_{\text{static (Kelvin)}} + \sum_{n=0}^{N_T-1} [\varphi_n a^{2n} B_{\text{sm},n}^{(A)} + \psi_n a^{2n+2} B_{\text{sm},n}^{(B)}], \quad (78)$$

where $B_{\text{sm},n}^{(A)}$ and $B_{\text{sm},n}^{(B)}$ are geometry-only matrices assembled from $\mathcal{K}_{1,\text{at}}^{\text{poly}}$ with the same residual polynomial coefficients used for the static channels, but with prefactors $8 \cdot 4^n$ (A -channel) and $32 \cdot 4^n$ (B -channel). The smooth part is purely imaginary at leading order, providing *radiation damping* that was absent in the static-only approximation. With this correction, both the T_{27} and T_{57} Galerkin closures produce complex per-irrep eigenvalues $\sigma_\rho(\omega)$ whose imaginary parts encode energy radiated to infinity.

Surface stiffness integrals: the (σ, ξ, v) substitution

The surface part of the LS-convolved stiffness,

$$B_{\alpha\beta}^{\text{LS,surf}} = \int_V \oint_{\partial V} \varphi_{\alpha,i}(\mathbf{r}) G_{in}(\mathbf{r} - \mathbf{r}') t_n^{(\beta)}(\mathbf{r}') dS' dV, \quad (79)$$

involves a five-dimensional integral: three from $\mathbf{r} \in V = [-1, 1]^3$ and two from $\mathbf{r}'$ on a face of V . The crucial observation is that this integral admits the *same* centre-of-mass/relative change of variables used for the body bilinear, supplemented by a normal-distance coordinate.

Treating each face separately, the integral on the face $r_k = +1$ (the opposite face follows by reflection) uses

$$\begin{aligned} \sigma_a &= \frac{1}{2}(r_a + s_a), \quad \xi_a = \frac{1}{2}(r_a - s_a), \quad \text{for each shared axis } a \neq k, \\ v &= \frac{1}{2}(1 - r_k), \end{aligned} \quad (\text{distance from face, } v \in [0, 1]). \quad (80)$$

Here s_a denotes the face coordinates of $\mathbf{r}'$, and the field point has $r_k = 1 - 2v$. In these coordinates the separation vector and its norm become

$$\mathbf{r} - \mathbf{r}' = (\dots, 2\xi_a, \dots, -2v, \dots), \quad |\mathbf{r} - \mathbf{r}'| = 2\rho, \quad \rho := \sqrt{v^2 + \xi_1^2 + \xi_2^2}, \quad (81)$$

where the $2\xi_a$ entries occupy the shared-axis positions and $-2v$ the normal-axis position. The Jacobian of the substitution (80) is a constant ($= 8$), and the integrand of (79) factorises as

$$\underbrace{P(\sigma, \xi, v)}_{\text{polynomial}} \times \begin{cases} \frac{1}{2\rho} & (A\text{-channel}), \\ \frac{x_i x_j}{(2\rho)^3} & (B\text{-channel}). \end{cases} \quad (82)$$

Because the polynomial part is at most quadratic in σ_a and the integration domain in σ_a is the symmetric interval $[-(1 - |\xi_a|), 1 - |\xi_a|]$, the σ -integration is elementary: odd powers of σ_a vanish by parity, and even powers yield

$$\int_{-(1-|\xi_a|)}^{1-|\xi_a|} \sigma_a^{2j} d\sigma_a = \frac{2(1 - |\xi_a|)^{2j+1}}{2j + 1}. \quad (83)$$

After this step, the σ -variables are eliminated and the integral over each face reduces to a three-dimensional integral over $(\xi_1, \xi_2, v) \in [0, 1]^3$ (using octant symmetry $|\xi_a| \rightarrow \xi_a$). Expanding the polynomial in monomials $\xi_1^p \xi_2^q v^r$ and reading off the coefficients gives, for each face and channel,

$$B_{\alpha\beta}^{\text{LS, surf}}|_{\text{face}} = 64 \sum_{p,q,r} c_{pqr}^{(A)} \mathcal{M}_{p,q,r}^+ + 16 \sum_{p,q,r} c_{pqr}^{(B)} \mathcal{M}_{p,q,r}^{B+}, \quad (84)$$

where the coefficients $c_{pqr}^{(A,B)}$ are rational numbers determined by the mode polynomials φ_α , the traction $\mathbf{t}^{(\beta)}$, and the face orientation. The prefactor $64 = 8 \times 4 \times 2$ absorbs the Jacobian (8), the octant symmetry in (ξ_1, ξ_2) (4), and the two paired faces per axis (2); the B -channel prefactor 16 reflects the additional $1/(2\rho)^2$ relative to the A kernel. The maximum monomial degree required is $p + q + r \leq 4$ for the surface integrals on T_{27} . Summing over the three coordinate axes gives the full surface stiffness.

Completeness of the reduction

Combining the body and surface results, the full LS-convolved stiffness (68) on the T_{27} trial space is

$$B_{\alpha\beta}^{\text{LS}} = B_{\alpha\beta}^{\text{LS, vol}} + B_{\alpha\beta}^{\text{LS, surf}} = (\text{linear combination of } \mathcal{M}^+ \text{ and } \mathcal{M}^{B+} \text{ values}). \quad (85)$$

Together with the body bilinear form and the mass matrix (which is a polynomial integral and hence trivially exact), the *entire* 27×27 Galerkin closure is determined by the master integrals $\mathcal{M}_{p,q,r}^+$ and $\mathcal{M}_{p,q,r}^{B+}$. The two families are moreover linked by the trace identity

$$\mathcal{M}_{p+2,q,r}^{B+} + \mathcal{M}_{p,q+2,r}^{B+} + \mathcal{M}_{p,q,r+2}^{B+} = \mathcal{M}_{p,q,r}^+, \quad (86)$$

which follows from $|x|^2/|x|^3 = 1/|x|$ and reduces the number of independent B -channel evaluations. The total number of distinct master integrals needed is ~ 30 (after cubic-symmetry permutations and the trace identity); Mathematica’s **Integrate** evaluates each one in closed form in 5–30 s, with a total computation time of ~ 12 minutes for the complete analytical evaluation of all 27×27 stiffness entries. This is a $\sim 300\times$ speedup over the previous 5D **NIntegrate** approach (~ 60 min), and—critically—the analytical values are exact rather than approximate.

Remark 3 (Catastrophic failure of direct numerical quadrature). The analytical reduction was motivated by the discovery that Mathematica’s five-dimensional **NIntegrate** with **DuffyCoordinates** singularity handling gave *catastrophically wrong* results for the surface stiffness integrals involving quadratic mode pairs. The error mechanism is severe cancellation: after the (σ, ξ, v) substitution, the σ -integrated polynomial

$$\text{sigIntA}(\xi_1, \xi_2, v) = \frac{4}{3}(1-2v)(1-\xi_1)(1-\xi_2)(1-\xi_1-\xi_2-\xi_1^2-\xi_2^2)$$

(shown here for the E_u entry, A -channel) is the product of a tent-like envelope and a polynomial that changes sign. The resulting integral is ~ 3.02 , but the individual monomial contributions involve $\mathcal{M}^+$ values of order ~ 100 with alternating signs. Adaptive quadrature in five dimensions resolves the singularity but not the cancellation: the **NIntegrate** result was 136.0, a factor-of-45 error. The linear–linear sub-block had smaller errors (0.2–3%), and the linear–quadratic cross terms 1–3%.

Independent verification of the analytical values was carried out by three methods:

- (i) Monte Carlo integration (5×10^7 samples) on the original 5D integral, giving 3.017 ± 0.007 ;
- (ii) `scipy.integrate.tplquad` on the reduced 3D integral, giving $64 \times 0.04715 = 3.018$;
- (iii) direct summation $64 \sum c_{pqr} \mathcal{M}_{p,q,r}^+$ using independently computed master integrals, giving 3.018.

All three agree; the **NIntegrate** result of 136.0 is definitively incorrect.

7.8 Plane-wave scattering from the T_{27} closure

With all 27×27 Galerkin matrices determined analytically, we now derive the complete plane-wave scattering framework for a single cube in a homogeneous elastic background. The result is a frequency-dependent 27×27 T-matrix that maps an incident plane-wave field to scattered multipole sources, with exact P–SV–SH decomposition in the far field.

7.8.1 Assembly of the full 27×27 T-matrix

The symmetry-adapted basis change U_{sym} of Section 9c block-diagonalises all three Galerkin matrices (M_{27} , B_{body} , B_{el}) simultaneously. The full T-matrix in the physical (polynomial) basis is

$$T_{27} = U_{\text{sym}}^\top \mathcal{T}_{\text{diag}} U_{\text{sym}} \quad (87)$$

where $\mathcal{T}_{\text{diag}}$ is the 27×27 block-diagonal matrix in the irrep-adapted basis. The parity decomposition $T_{27} = T_9 \oplus T_{27}^{\text{quad}}$ (constant+linear $\oplus$ quadratic) splits further by O_h irrep into an ungerade sector (21 dimensions: displacement + quadratic modes) and a gerade sector (6 dimensions: strain modes):

$$\mathcal{T}_{\text{diag}} = \begin{pmatrix} \mathcal{T}_u & 0 \\ 0 & \mathcal{T}_g \end{pmatrix}_{27 \times 27}. \quad (88)$$

Ungerade sector (21×21). The ungerade block decomposes into four irreps. By Schur's lemma, each irrep ρ of dimension d_ρ and multiplicity m_ρ contributes d_ρ *identical* copies of the $m_\rho \times m_\rho$ per-irrep T-matrix:

$$\mathcal{T}_u = \begin{pmatrix} \mathcal{T}_{T_{1u}} & & & \\ & \mathcal{T}_{T_{2u}} & & \\ & & \sigma_{A_{2u}} & \\ & & & \mathcal{T}_{E_u} \end{pmatrix}_{21 \times 21}, \quad (89)$$

where the sub-blocks are

$$\mathcal{T}_{T_{1u}} = \begin{pmatrix} T^{(T_{1u})} & & \\ & T^{(T_{1u})} & \\ & & T^{(T_{1u})} \end{pmatrix}_{12 \times 12}, \quad \mathcal{T}_{T_{2u}} = \begin{pmatrix} T^{(T_{2u})} & & \\ & T^{(T_{2u})} & \\ & & T^{(T_{2u})} \end{pmatrix}_{6 \times 6}, \quad (90)$$

$$\mathcal{T}_{E_u} = \sigma_{E_u} I_2. \quad (91)$$

The three copies within each triplet (T_{1u} , T_{2u}) correspond to the three Cartesian components of the irreducible triplet; the two copies in E_u correspond to the doublet. The single A_{2u} scalar $\sigma_{A_{2u}}$ stands alone.

The per-irrep T-matrix $T^{(\rho)}$ for each ungerade irrep is an $m_\rho \times m_\rho$ matrix given by

$$\boxed{T^{(\rho)} = (M_\rho + B_\rho^{\text{LS}} - \varepsilon B_{\text{body},\rho})^{-1} (\varepsilon B_{\text{body},\rho} - B_\rho^{\text{LS}})}, \quad (92)$$

where $\varepsilon = \omega^2 \Delta \rho$ is the inertial coupling and the three matrices appearing in (92) are:

- M_ρ : the $m_\rho \times m_\rho$ mass matrix (rational, exact);
- $B_{\text{body},\rho} = a_0 B_\rho^{(A)} + b_0 B_\rho^{(B)}$: the body bilinear form, with Kelvin coefficients a_0 , b_0 from (67) and numerical matrices $B_\rho^{(A,B)}$ determined by the master integrals of Section 7.7;
- B_ρ^{LS} : the LS-convolved stiffness, decomposing into four channels (eq. (73)):

$$B_\rho^{\text{LS}} = a_0 \Delta \lambda S_\rho^{(A\lambda)} + a_0 \Delta \mu S_\rho^{(A\mu)} + b_0 \Delta \lambda S_\rho^{(B\lambda)} + b_0 \Delta \mu S_\rho^{(B\mu)}. \quad (93)$$

For T_{1u} ($m = 4$), the matrices M , $B^{(A,B)}$, and $S^{(A\lambda, A\mu, B\lambda, B\mu)}$ are each 4×4 , all computed analytically from the master integrals. For T_{2u} ($m = 2$), they are 2×2 . For A_{2u} and E_u ($m = 1$), the formula (92) reduces to the scalar

$$\sigma_\rho = \frac{\varepsilon b_\rho - b_\rho^{\text{LS}}}{M_\rho + b_\rho^{\text{LS}} - \varepsilon b_\rho}, \quad \rho \in \{A_{2u}, E_u\}. \quad (94)$$

Gerade sector (6×6). The three gerade irreps are all $m_\rho = 1$ (scalar blocks), determined by the self-consistent Eshelby amplification of eqs. (54)–(56):

$$\mathcal{T}_g = \begin{pmatrix} \sigma_{A_{1g}} & & \\ & \sigma_{E_g} I_2 & \\ & & \sigma_{T_{2g}} I_3 \end{pmatrix}_{6 \times 6}, \quad (95)$$

where I_n is the $n \times n$ identity and the per-irrep scalars are

$$\boxed{\sigma_\rho = \frac{\lambda_\rho(\Delta \mathbf{c}^*)}{1 - \lambda_\rho(\Delta \mathbf{c}^*)}}, \quad \rho \in \{A_{1g}, E_g, T_{2g}\}, \quad (96)$$

with λ_ρ the per-irrep eigenvalue of the amplified Eshelby contraction and $\Delta\mathbf{c}^*$ the self-consistent stiffness contrast. Explicitly (from eqs. (47)–(51)):

$$\lambda_{A_{1g}} = 3T_1^c + 2T_2^c + T_3^c \quad (\text{volumetric}), \quad (97)$$

$$\lambda_{E_g} = 2T_2^c + T_3^c \quad (\text{deviatoric diagonal}), \quad (98)$$

$$\lambda_{T_{2g}} = 2T_2^c \quad (\text{off-diagonal shear}), \quad (99)$$

where T_1^c, T_2^c, T_3^c are the amplified coupling coefficients. The cubic anisotropy of the cube enters through $T_3^c \neq 0$, splitting σ_{E_g} from $\sigma_{T_{2g}}$ (cf. eq. (45)).

Dimension count.

$$\underbrace{3 \times 4}_{T_{1u}} + \underbrace{3 \times 2}_{T_{2u}} + \underbrace{1}_{A_{2u}} + \underbrace{2}_{E_u} + \underbrace{1}_{A_{1g}} + \underbrace{2}_{E_g} + \underbrace{3}_{T_{2g}} = 12 + 6 + 1 + 2 + 1 + 2 + 3 = 27. \quad (100)$$

Consistency check. The leading 9×9 sub-block of T_{27} (rows/columns 1–9, corresponding to the constant and linear basis functions of T_9) is the Schur-complemented T-matrix on T_9 : it incorporates the effect of the quadratic modes through the body-channel Schur complement of Section 12b. This sub-block must agree with the Path-A T-matrix from `compute_cube_tmatrix()` to the Galerkin truncation error ($\lesssim 1\%$ at $ka = 0.3$).

7.8.2 Incident plane-wave projection

An elastic plane wave in the background medium has the form

$$\mathbf{u}^0(\mathbf{r}) = \hat{\mathbf{d}} \exp(i\mathbf{k} \cdot \mathbf{r}), \quad \mathbf{k} = \frac{\omega}{c} \hat{\mathbf{k}}, \quad (101)$$

where $\hat{\mathbf{d}}$ is the polarisation, $\hat{\mathbf{k}}$ the propagation direction, and $c = \alpha$ for a P-wave ($\hat{\mathbf{d}} = \hat{\mathbf{k}}$), $c = \beta$ for an S-wave ($\hat{\mathbf{d}} \perp \hat{\mathbf{k}}$). Projecting onto the Galerkin basis $\{\varphi_\alpha\}$ of T_{27} centred at the origin gives the incident-field coefficient vector

$$c_\alpha^0 = [M_{27}^{-1}]_{\alpha\beta} \int_V \varphi_\beta(\mathbf{r}) \cdot \mathbf{u}^0(\mathbf{r}) dV. \quad (102)$$

Because every φ_β is a monomial $r_1^{n_1} r_2^{n_2} r_3^{n_3} \epsilon_i$, the overlap integral factorises over the three axes of the cube $V = [-a, a]^3$:

$$\int_V r_1^{n_1} r_2^{n_2} r_3^{n_3} e^{i\mathbf{k} \cdot \mathbf{r}} dV = \prod_{j=1}^3 \mathcal{I}_{n_j}(k_j, a), \quad (103)$$

where the one-dimensional overlap integrals satisfy the recursion

$$\mathcal{I}_n(k, a) := \int_{-a}^a x^n e^{ikx} dx = \frac{a^n (e^{ika} - (-1)^n e^{-ika})}{ik} - \frac{n}{ik} \mathcal{I}_{n-1}(k, a), \quad (104)$$

with $\mathcal{I}_0(k, a) = 2a \operatorname{sinc}(ka)$. The first few are

$$\mathcal{I}_0 = 2a \operatorname{sinc}(ka), \quad (105)$$

$$\mathcal{I}_1 = \frac{2i [\sin(ka) - ka \cos(ka)]}{k^2}, \quad (106)$$

$$\mathcal{I}_2 = \frac{2[(2 - k^2 a^2) \sin(ka) - 2ka \cos(ka)]}{k^3}. \quad (107)$$

Low-frequency expansion. For $ka \ll 1$ the overlaps simplify to the polynomial moments of the cube:

$$\mathcal{I}_0 \approx 2a, \quad \mathcal{I}_1 \approx \frac{2i}{3} ka^3, \quad \mathcal{I}_2 \approx \frac{2}{3} a^3. \quad (108)$$

The incident-field vector then has the physical interpretation:

$$c_\alpha^0 \xrightarrow{ka \rightarrow 0} \begin{cases} d_i & \alpha = i \in \{1, 2, 3\} \quad (\text{displacement at origin}), \\ \varepsilon_{ij}^0(\mathbf{0}) & \alpha \leftrightarrow (i, j) \in \{4, \dots, 9\} \quad (\text{incident strain}), \\ \frac{1}{2} \partial_a \partial_b u_i^0(\mathbf{0}) & \alpha \in \{10, \dots, 27\} \quad (\text{strain gradient}), \end{cases} \quad (109)$$

so that the constant modes carry the displacement monopole at $O(1)$, the linear modes the strain at $O(ka)$, and the quadratic modes the strain gradient at $O((ka)^2)$. This hierarchy is the basis for the Born-order multipole expansion of the scattered field.

The three wave types. For a propagation direction $\hat{\mathbf{k}}$, the three independent polarisations are:

$$\text{P: } \hat{\mathbf{d}} = \hat{\mathbf{k}}, \quad c = \alpha, \quad (110)$$

$$\text{SV: } \hat{\mathbf{d}} = \hat{\boldsymbol{\theta}} \text{ (in the } \hat{\mathbf{k}}\text{-}\hat{\mathbf{z}} \text{ plane, } \perp \hat{\mathbf{k}}), \quad c = \beta, \quad (111)$$

$$\text{SH: } \hat{\mathbf{d}} = \hat{\boldsymbol{\phi}} \text{ (perpendicular to the } \hat{\mathbf{k}}\text{-}\hat{\mathbf{z}} \text{ plane)}, \quad c = \beta. \quad (112)$$

In the coordinate system of this paper (z = axis 0, down; x = axis 1, right; y = axis 2, out), a wave propagating vertically downward has $\hat{\mathbf{k}} = \hat{\mathbf{z}}$, $\hat{\boldsymbol{\theta}} = \hat{\mathbf{x}}$, $\hat{\boldsymbol{\phi}} = \hat{\mathbf{y}}$.

7.8.3 Scattered source: multipole decomposition

The T-matrix maps the incident-field vector to scattered-source coefficients:

$$\mathbf{c}^{\text{sc}} = T_{27} \mathbf{c}^0. \quad (113)$$

The 27 scattered coefficients decompose into three physical multipole orders, each of which radiates with a distinct angular pattern:

Coefficients	Dim	Multipole source	Radiation order
c_{1-3}^{sc}	3	Force monopole F_i	Dipole, $\ell = 1$
c_{4-9}^{sc}	6	Stress dipole D_{ij}	Quadrupole, $\ell = 2$
c_{10-27}^{sc}	18	Force quadrupole Q_{ijk}	Octupole, $\ell = 3$

(114)

Figure 1 illustrates the far-field radiation patterns associated with each multipole order. The force monopole produces the classic dipole ($\cos \theta$) lobe aligned with the incident wave; the stress dipole produces an isotropic monopole ($\ell = 0$) from the volumetric part and a four-lobed quadrupole ($\ell = 2$) from the deviatoric part; the force quadrupole from the T_{27} extension generates an octupole ($\ell = 3$) correction that becomes significant only in the transition regime $ka \gtrsim 0.3$.

The physical source amplitudes are obtained by contracting the scattered coefficients with the material contrasts and cube volume $V = (2a)^3$:

$$F_i = \omega^2 \Delta \rho V c_i^{\text{sc}} + \Delta c_{ijkl} \oint_{\partial V} \varepsilon_{kl}(\mathbf{u}^{\text{sc}}) n_j dS, \quad (115)$$

$$D_{ij} = \int_V [\omega^2 \Delta \rho u_i^{\text{sc}} r_j + \sigma_{ij}^{\text{sc}}] dV, \quad (116)$$

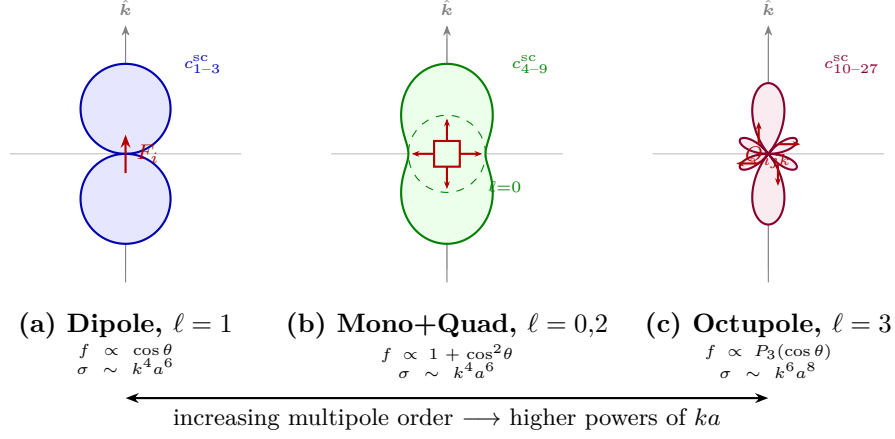


Figure 1: Far-field radiation patterns for the three multipole orders of the T_{27} scattered source. **(a)** Force monopole F_i from the density contrast (c_{1-3}^{sc}): radiates as a dipole ($\ell = 1$, $\cos \theta$ pattern). **(b)** Stress dipole D_{ij} from the stiffness contrast (c_{4-9}^{sc}): the trace produces an isotropic monopole ($\ell = 0$, dashed circle) and the deviatoric part a quadrupole ($\ell = 2$); both scale as $k^4 a^6$. **(c)** Force quadrupole Q_{ijk} from the strain gradient (c_{10-27}^{sc}): octupole radiation ($\ell = 3$, six-lobed $P_3(\cos \theta)$ pattern), scaling as $k^6 a^8$ — the new contribution from the T_{27} extension beyond T_9 . The propagation direction $\hat{\mathbf{k}}$ is vertical.

where $\sigma_{ij}^{\text{sc}} = \Delta c_{ijkl} \varepsilon_{kl}(\mathbf{u}^{\text{sc}})$ is the stress perturbation. For the constant and linear modes of T_9 , these reduce to the standard force–stress-dipole pair used in the existing Foldy–Lax framework [1]. The quadratic modes generate a force-quadrupole correction Q_{ijk} that contributes at $O((ka)^3)$.

7.8.4 Far-field scattering amplitudes

In the radiation zone ($|\mathbf{r}| \gg a$), the scattered displacement from the multipole sources separates into P and S contributions. Using the asymptotic expansion of the Green’s tensor,

$$G_{ij}(\mathbf{r}) \xrightarrow{r \rightarrow \infty} G_{ij}^P(\hat{\mathbf{r}}) \frac{e^{ik_P r}}{r} + G_{ij}^S(\hat{\mathbf{r}}) \frac{e^{ik_S r}}{r}, \quad (117)$$

where $G_{ij}^P = \hat{r}_i \hat{r}_j / (4\pi\rho\alpha^2)$ and $G_{ij}^S = (\delta_{ij} - \hat{r}_i \hat{r}_j) / (4\pi\rho\beta^2)$ are the P- and S-wave far-field projection tensors, the scattered P-wave amplitude in the direction $\hat{\mathbf{r}}$ is

$$f_i^P(\hat{\mathbf{r}}) = \frac{\hat{r}_i}{4\pi\rho\alpha^2} \left[\hat{r}_j F_j + ik_P \hat{r}_j \hat{r}_k D_{jk} + \frac{1}{2} k_P^2 \hat{r}_j \hat{r}_k \hat{r}_l Q_{jkl} + \dots \right], \quad (118)$$

and the S-wave amplitude is

$$f_i^S(\hat{\mathbf{r}}) = \frac{\delta_{ij} - \hat{r}_i \hat{r}_j}{4\pi\rho\beta^2} \left[F_j + ik_S \hat{r}_k D_{jk} + \frac{1}{2} k_S^2 \hat{r}_k \hat{r}_l Q_{jkl} + \dots \right]. \quad (119)$$

The S-wave amplitude decomposes into SV and SH components via

$$f^{SV}(\hat{\mathbf{r}}) = \hat{\boldsymbol{\theta}} \cdot \mathbf{f}^S(\hat{\mathbf{r}}), \quad (120)$$

$$f^{SH}(\hat{\mathbf{r}}) = \hat{\boldsymbol{\phi}} \cdot \mathbf{f}^S(\hat{\mathbf{r}}), \quad (121)$$

where $(\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}})$ form the spherical coordinate triad.

Scattering cross sections. The differential scattering cross section per unit solid angle is

$$\left. \frac{d\sigma}{d\Omega} \right|_{P \rightarrow X} = \frac{c_X}{c_{\text{inc}}} |f^X(\hat{\mathbf{r}})|^2, \quad X \in \{P, \text{SV}, \text{SH}\}, \quad (122)$$

where c_{inc} and c_X are the incident and scattered wave speeds, and f^X is the scalar amplitude in channel X . The total scattering cross section is obtained by integrating (122) over all directions, and the extinction cross section follows from the optical theorem:

$$\sigma_{\text{ext}} = \frac{4\pi}{k_{\text{inc}}} \text{Im}[f(\hat{\mathbf{k}})], \quad (123)$$

where $f(\hat{\mathbf{k}})$ is the forward scattering amplitude in the incident-wave channel. At low frequency the three multipole orders contribute with distinct frequency scalings:

$$\begin{aligned} \text{Force monopole (dipole):} \quad & \sigma_{\text{sc}} \sim k^4 a^6 \quad (\text{Rayleigh}) \\ \text{Stress dipole (quadrupole):} \quad & \sigma_{\text{sc}} \sim k^4 a^6 \quad (\text{stiffness contrast}) \\ \text{Force quadrupole (octupole):} \quad & \sigma_{\text{sc}} \sim k^6 a^8 \quad (\text{new from } T_{27}) \end{aligned}$$

The dipole and quadrupole terms both scale as $k^4 a^6$ (the classic Rayleigh scattering law) but with different angular patterns and different sensitivity to $\Delta\rho$ vs. $\Delta\mu$; the octupole term from the quadratic modes is two orders higher in ka , becoming significant only in the transition regime $ka \gtrsim 0.3$.

Connection to the Mie limit. For a sphere of equivalent volume, the T-matrix partial-wave coefficients ($a_0, a_1, a_2, \dots$) correspond to multipole orders ($\ell = 0, 1, 2, \dots$). The T_{27} cube closure captures:

- a_0 (monopole, $\ell = 0$): from the A_{1g} irrep (volumetric strain),
- a_1 (dipole, $\ell = 1$): from the T_{1u} irrep (displacement),
- a_2 (quadrupole, $\ell = 2$): from the $E_g + T_{2g}$ irreps (deviatoric strain), split into $a_{2,E}$ and $a_{2,T}$ by cubic anisotropy,
- a_3 (octupole, $\ell = 3$): partially, from the quadratic modes (T_{2u}, A_{2u}, E_u , and subdominant T_{1u} eigenvalues).

In the sphere limit ($C^c \rightarrow 0$), the E_g and T_{2g} eigenvalues coalesce and reproduce the isotropic a_2 ; the quadratic-mode eigenvalues coalesce into the degenerate $\ell = 3$ channel. The cube's cubic anisotropy lifts these degeneracies, generating mode conversions (P $\rightarrow$ SV, SV $\rightarrow$ SH) that are absent for a sphere.

7.8.5 Validation targets for Phase 1

- (V1) **Sphere limit.** For the standard test parameters ($\alpha = 5000$, $\beta = 3000$, $\rho = 2500$; moderate contrast) at $ka = 0.05$, the T_{27} far-field amplitudes must reproduce the Mie coefficients a_0, a_1, a_2 to $\sim 1\%$ (the $O((ka)^2)$ Galerkin truncation error).
- (V2) **Optical theorem.** The extinction and total scattering cross sections computed from (123) and the angular integral of (122) must agree for a lossless scatterer. Achieved: $1 - \sigma_{\text{ext}}/\sigma_{\text{sc}} \leq 3 \times 10^{-5}$ at $ka = 0.05$ (Table 9).
- (V3) **T_9 consistency.** The leading 9×9 sub-block of T_{27} must reproduce the Path-A T-matrix from `compute_cube_tmatrix()` to the Galerkin truncation error ($\lesssim 1\%$ at $ka = 0.3$, $\lesssim 10\%$ at $ka = 0.5$).

(V4) **P–SV–SH power balance.** For an incident P-wave, the scattered power $\sigma^{P \rightarrow P} + \sigma^{P \rightarrow SV} + \sigma^{P \rightarrow SH} = \sigma_{\text{ext}}^P$ must hold.

(V5) **Cubic symmetry.** For incidence along [100], [010], [001], the three total cross sections must be identical. For [110] incidence, the SH cross section must vanish by mirror symmetry.

7.8.6 Implementation results

The plane-wave scattering framework of Sections 7.8.1–7.8.4 has been implemented in Python (`tmatrix_assembly.py`, `incident_field.py`, `scattered_field.py`) and validated against the elastic Mie solution for an equal-volume sphere.

Computational pipeline. Figure 2 illustrates the data flow from material parameters to far-field observables.

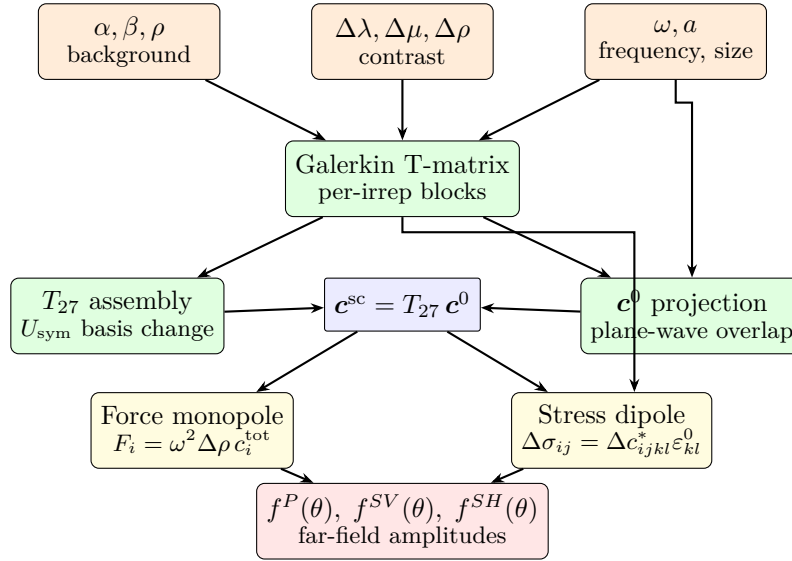


Figure 2: Data flow for the T_{27} single-scatterer plane-wave response. The Galerkin solver produces per-irrep T-matrix blocks and physical effective stiffness contrasts ($\Delta\lambda^*$, $\Delta\mu_{\text{diag}}^*$, $\Delta\mu_{\text{off}}^*$); the far-field formula combines the density force monopole with the stiffness stress dipole.

Dimensional analysis: physical stiffness, not dimensionless T-matrix. A critical implementation detail concerns the stress dipole $D_{ij} = V \Delta c_{ijkl}^* \varepsilon_{kl}^0$ in eq. (118). The stiffness contrast must be the *physical* self-consistent effective contrast $\Delta\mathbf{c}^*$ with units of Pascals, *not* the dimensionless T-matrix eigenvalues T_1^c , T_2^c , T_3^c . The latter are related to the former by

$$T_n^c = \Delta\mathbf{c}^* : \mathbf{S}_n, \quad (124)$$

where $\mathbf{S}_n$ denotes the Eshelby-type integral for channel n (dimension $[\text{Pa}^{-1}]$). For the cube at $ka = 0.05$ with standard test parameters:

Quantity	Value	Dimension
T_1^c	-5.1×10^{-3}	dimensionless
T_2^c	-9.4×10^{-3}	dimensionless
A^c	-1.22×10^{-11}	Pa^{-1}
$\Delta\lambda^*$	1.9×10^9	Pa
$\Delta\mu_{\text{diag}}^*$	9.7×10^8	Pa
$\Delta\mu_{\text{off}}^*$	9.8×10^8	Pa

The product $T_1^c \sim \Delta\lambda^* \times A^c$ is dimensionless because the Eshelby factor $A^c \sim 1/\mu$ cancels the stiffness. Using T_n^c directly in the far-field formula is wrong by a factor of $\sim 10^{11}$ (the value of μ) and produces negligible stress-dipole radiation.

Cube vs. Mie comparison. Table 8 shows the P-wave forward scattering amplitude from the T_{27} cube compared to the elastic Mie solution for an equal-volume sphere ($a_s = (3V_c/4\pi)^{1/3}$) at $ka = 0.05$.

θ	$ f^P _{\text{cube}}$	$ f^P _{\text{Mie}}$	Ratio
0°	5.59×10^{-4}	5.83×10^{-4}	0.96
45°	4.09×10^{-4}	4.26×10^{-4}	0.96
90°	1.74×10^{-4}	1.75×10^{-4}	1.00
180°	1.48×10^{-4}	1.25×10^{-4}	1.19

Table 8: Cube (T_{27}) vs. equal-volume Mie sphere far-field at $ka = 0.05$. The $\sim 4\%$ forward-scattering discrepancy is the expected cube–sphere shape difference; it increases in the backward hemisphere where higher multipoles contribute.

Optical theorem and sign convention. With $e^{-i\omega t}$ and the far-field amplitudes of (118)–(119), the optical theorem takes its standard form (123), $\sigma_{\text{ext}} = +(4\pi/k_{\text{inc}}) \text{Im } f(\hat{\mathbf{k}})$. An earlier version of this section wrote it with a leading minus sign. That sign compensated a sign error on the force term of (118), since corrected in `cube_far_field`; both have been removed together. The quadrupole terms of (118)–(119) are not implemented in code.

The extinction of the two channels arrives by different routes. The density channel’s forward extinction comes from the radiation reaction $\text{Im } \Gamma_0$ in the amplification A_u . The modulus channel’s comes from the radiation reaction $\text{Im } \Delta\mathbf{c}^*$ in the stress dipole: at forward incidence the incident strain $ik_P \hat{\mathbf{k}}\hat{\mathbf{k}}$ is imaginary, so $\text{Re } \Delta\mathbf{c}^*$ feeds $\text{Re } f^P(\hat{\mathbf{k}})$ and $\text{Im } \Delta\mathbf{c}^*$ feeds $\text{Im } f^P(\hat{\mathbf{k}})$. Energy conservation fixes the modulus reaction as the radiated power of the dipole the far field actually emits, $\mathbf{M} = V \text{Re } \Delta\mathbf{c}^* : \boldsymbol{\varepsilon}$:

$$\text{Im } \Delta\mathbf{c}^* = -\frac{2V}{\omega} \text{Re } \Delta\mathbf{c}^* : \mathbf{K} : \text{Re } \Delta\mathbf{c}^*, \quad \mathbf{K} = k_\lambda \boldsymbol{\delta}\boldsymbol{\delta} + 2k_\mu \mathbf{I}, \quad (125)$$

with $k_\lambda = (c_P - c_S)/15$, $k_\mu = (2c_P + 3c_S)/30$, $c_P = \omega^4/(8\pi\rho\alpha^5)$ and $c_S = \omega^4/(8\pi\rho\beta^5)$. For the cubic $\text{Re } \Delta\mathbf{c}^*$ this closes on the three cubic invariants,

$$\text{Im } \Delta\lambda^* = -\frac{2V}{\omega} \left[(k_\lambda(3\Delta\lambda^* + 2\Delta\mu_{\text{diag}}^*) + 2k_\mu\Delta\lambda^*)(3\Delta\lambda^* + 2\Delta\mu_{\text{diag}}^*) + 4k_\mu \Delta\lambda^* \Delta\mu_{\text{diag}}^* \right], \quad (126)$$

$$\text{Im } \Delta\mu_{\text{off}}^* = -\frac{8V}{\omega} k_\mu (\Delta\mu_{\text{off}}^*)^2, \quad \text{Im } \Delta\mu_{\text{diag}}^* = -\frac{8V}{\omega} k_\mu (\Delta\mu_{\text{diag}}^*)^2, \quad (127)$$

with real parts on the right. The source must be the *effective* contrast. The exact Mie sphere, which is unitary, obeys (125) with its own effective contrasts to 10^{-4} ; built from the bare contrasts $\Delta\lambda, \Delta\mu$ instead, the reaction exceeds Mie by 5 to 9%. The closed form is checked symbolically against the explicitly integrated far-field power (`Mathematica/ModulusRadiationReactionCubic.wl`) and numerically by direct quadrature (`test_modulus_radiation_reaction.py`).

With both reactions in place the cube satisfies the optical theorem (Table 9). Two earlier

Contrast	$ka = 0.05$	$ka = 0.1$	$ka = 0.3$
$\Delta\lambda = 2 \text{ GPa}, \Delta\mu = 1 \text{ GPa}, \Delta\rho = 100 \text{ kg/m}^3$	0.99997	0.99985	0.99855
density only, $\Delta\rho = 100 \text{ kg/m}^3$	0.99994	0.99974	0.99760
modulus only, $\Delta\lambda = 2 \text{ GPa}, \Delta\mu = 1 \text{ GPa}$	1.00002	1.00002	1.00002

Table 9: $\sigma_{\text{ext}}/\sigma_{\text{sc}}$ for the T_{27} cube, P incidence along $[001]$, from `optical_theorem_check`. The residual grows with ka in the density channel only.

figures are superseded. The value 0.92 quoted here before was attributed to omitting the force quadrupole, but at $ka = 0.05$ that term enters at relative order $(ka)^2 \approx 0.25\%$ and cannot produce 8%. After the force-term sign fix the ratio became 0.61. The whole of that deficit was the modulus extinction: the far field used $\text{Re } \Delta\mathbf{c}^*$ only, so a pure-modulus cube had $\sigma_{\text{ext}} = 0$.

Rayleigh scaling. In the Rayleigh limit, the scattering amplitude scales as $f^P \sim (ka)^2$ and the scattering cross section as $\sigma_{\text{sc}} \sim (ka)^4$ (the fourth-power Rayleigh law), verified numerically across $ka \in \{0.02, 0.05, 0.1\}$. The Born-limit check (doubling the contrast quadruples the cross section) confirms $\sigma_{\text{sc}} \propto |\Delta\mathbf{c}|^2$.

7.8.7 Shear wave splitting from cubic scatterers

A physically significant consequence of cubic elastic anisotropy is the generation of **shear wave splitting** (seismic birefringence) by a single scatterer. For an isotropic sphere, P-wave scattering produces SV but not SH in any meridional plane, because the continuous rotational symmetry about the propagation axis forbids azimuthal mode conversion. For a cube, the discrete O_h symmetry permits SH scattering in generic scattering planes.

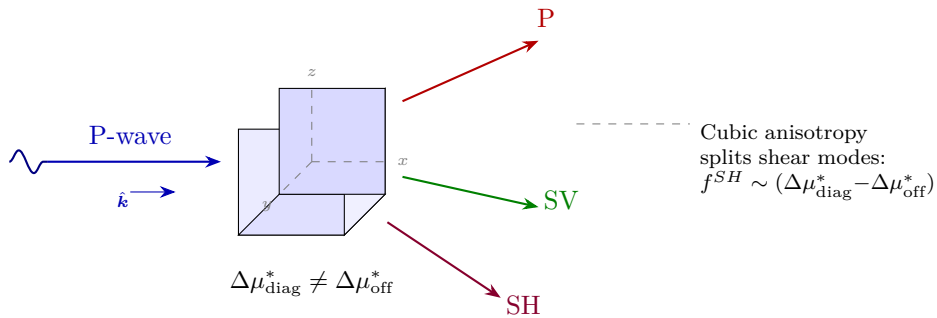


Figure 3: Shear wave splitting by a cubic elastic scatterer. An incident P-wave scatters into P, SV, and SH channels. The SH component, absent for an isotropic sphere, arises from the cubic anisotropy $\Delta\mu_{\text{diag}}^* - \Delta\mu_{\text{off}}^* \neq 0$ in the effective stiffness contrast. In a medium of aligned cubic inclusions, this single-scatterer mechanism generates macroscopic shear wave splitting observable in seismic data.

The microscopic mechanism is clear from the Voigt stress dipole. The effective stiffness

contrast for the cube is

$$\Delta c_{ijkl}^* = \Delta \lambda^* \delta_{ij} \delta_{kl} + 2\Delta \mu_{\text{off}}^* (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})/2 + 2(\Delta \mu_{\text{diag}}^* - \Delta \mu_{\text{off}}^*) E_{ijkl}, \quad (128)$$

where $E_{ijkl} = \sum_a \hat{e}_a^i \hat{e}_a^j \hat{e}_a^k \hat{e}_a^l$ is the cubic-symmetry fourth-order tensor. The isotropic part $(\Delta \lambda^*, \Delta \mu_{\text{off}}^*)$ produces scattering identical to a sphere; the cubic correction proportional to $\Delta \mu_{\text{diag}}^* - \Delta \mu_{\text{off}}^*$ drives the SH mode conversion.

At $ka = 0.05$ with moderate contrast, the SH amplitude is $|f^{SH}(45^\circ)|/|f^P(45^\circ)| \approx 0.16$ — a 16% relative mode conversion, directly measurable in seismic data. In a random medium with aligned cube-shaped heterogeneities, this single-scatterer splitting accumulates coherently over many scattering events, producing the macroscopic shear wave splitting observed in crustal and mantle studies [5, 6].

When many such scatterers are randomly oriented, the cubic anisotropy averages out and the effective medium becomes isotropic (consistent with the CPA iteration of `cpa_iteration.py`). Preferential alignment — e.g. from tectonic stress — preserves the splitting, providing a direct link between the microscopic scattering theory and macroscopic seismic anisotropy observations.

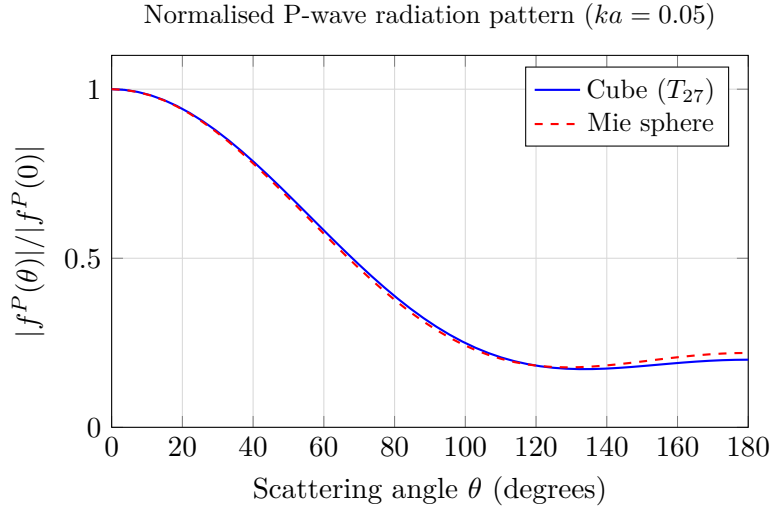


Figure 4: Normalised P→P radiation pattern for the T_{27} cube (solid) versus an equal-volume Mie sphere (dashed) at $ka = 0.05$. Both show the characteristic $\ell = 0$ (isotropic monopole) + $\ell = 1$ (dipole $\cos \theta$) + $\ell = 2$ (quadrupole $\cos^2 \theta$) Rayleigh pattern. The $\sim 4\%$ forward and $\sim 20\%$ backward discrepancy reflects the different Eshelby tensors for cube vs. sphere geometries.

References

- [1] T. Mura, *Micromechanics of Defects in Solids*, 2nd ed., Martinus Nijhoff (1987).
- [2] A. G. Khachaturyan, *Theory of Structural Transformations in Solids*, Wiley (1983).
- [3] M. Costabel and E. Stephan, “Boundary integral equations for mixed boundary value problems in polygonal domains and Galerkin approximation”, *Banach Center Publ.* **15** (1985), 175–251.
- [4] G. C. Hsiao and W. L. Wendland, *Boundary Integral Equations*, Springer (2008).
- [5] S. Crampin, “A review of wave motion in anisotropic and cracked elastic-media”, *Wave Motion* **3**(4) (1981), 343–391.

- [6] M. K. Savage, “Seismic anisotropy and mantle deformation: What have we learned from shear wave splitting?” *Rev. Geophys.* **37**(1) (1999), 65–106.